arXiv:2606.20818v2 [math.FA] 23 Aug 2026

WEYL CURVES AND ZETA DETERMINANTS OF CONIC LAPLACIANS ON RIEMANN SURFACES

JIA-MING (FRANK) LIOU

Abstract. We study self-adjoint realizations of conic Laplacians on compact Riemann surfaces with radial conic metrics whose cone angles are integral multiples of 2π. Their critical asymptotic coefficients span a finite-dimensional space with a nondegenerate skew-Hermitian Green form, whose Lagrangian subspaces parametrize the self-adjoint realizations. We construct the associated Weyl functions and derive Kreĭn's resolvent formula, a resolvent trace identity, and, under explicit zeta-regularity hypotheses, a comparison formula for positive-spectrum zeta determinants. The boundary data of formal solutions define a holomorphic Weyl curve in the Grassmannian, with the Weyl functions as its local graph coordinates. Its real restriction is a positive curve in the Lagrangian Grassmannian, and its tangent form is identified with the L^2-inner product through the Poisson operator. We further identify the boundary determinants with the transition functions of the determinant line bundle induced by the Weyl curve.

Keywords. Conic Laplacians; self-adjoint extensions; symplectic boundary data; Lagrangian Grassmannians; Weyl functions; Weyl curves; Kreĭn's resolvent formula; positive-spectrum zeta determinants; determinant line bundles.

1. Introduction

The study of elliptic operators on spaces with conic singularities has a long history, tracing back to Weyl's limit-point/limit-circle analysis of singular Sturm–Liouville operators and to the abstract theory of self-adjoint extensions of symmetric operators; see [15, 21]. In geometric analysis, conic singularities have been studied through the spectral theory of singular Riemannian spaces, beginning with Cheeger's work, and through the analysis of Fuchs-type differential operators and elliptic cone operators; see, for example, [4, 5, 17, 14]. In these theories, the leading asymptotic behavior of functions near the singular set determines a finite-dimensional symplectic boundary space, and self-adjoint realizations are described by Lagrangian boundary conditions on this space; see [8]. The geometry and spectra of closed extensions of elliptic cone operators were studied from a Grassmannian viewpoint by Gil, Krainer, and Mendoza [9].

In extension theory, Kreĭn-type formulas express resolvent differences in terms of Weyl functions; see [6]. For elliptic boundary value problems, these functions often appear as Dirichlet-to-Neumann type operators [2, 3]. Zeta determinants on conic manifolds were studied in [11, 12], while comparison, gluing, and relative determinant formulas were developed in [7, 1, 16]. Determinant lines over Grassmannians of elliptic boundary conditions and their relation to zeta and boundary Fredholm determinants were studied by Scott and Wojciechowski [19, 20]; see also [18].

The work most closely related to the present paper is that of Hillairet and Kokotov [10], who studied Euclidean surfaces with conical singularities and used Kreĭn's formula, together with the associated S-matrix, to derive comparison formulas for the zeta-regularized determinants of non-Friedrichs self-adjoint extensions of the Laplacian. Our approach is complementary: rather than using the S-matrix as the main boundary object, we formulate the resolvent and determinant comparison problems in terms of a finite-dimensional boundary symplectic space and its Weyl functions.

We consider radial conic metrics with cone angles $2\pi n_P$, where each n_P is an integer. The critical asymptotic coefficients at the conic points identify $D_{\max}(\Delta_X)/D_{\min}(\Delta_X)$ with a finite-dimensional space V_S, on which the Green form induces a nondegenerate skew-Hermitian form ω_S. We give a direct Fourier-mode construction that fixes the normalization needed later. Self-adjoint realizations are then parametrized by Lagrangian subspaces of (V_S, ω_S). For every Lagrangian pair, we construct the associated Poisson operator and Weyl function and derive a finite-rank Kreĭn resolvent formula. Its trace yields the logarithmic derivative of a finite-dimensional boundary determinant $d^X_{\mathcal{V},\mathcal{W}}$. This leads to a comparison formula for positive-spectrum zeta determinants, including the correction from the omitted nonpositive spectrum. Since general self-adjoint conic realizations may have spectral zeta functions with nonstandard singularities, the determinant comparison is stated for pairs of realizations satisfying the explicit zeta-regularity hypothesis in Section 5.

Although the extension-theoretic parametrization, the abstract Kreĭn formula, and the general notion of a Weyl function are standard, their explicit normalized realization for conic critical asymptotic data allows the same boundary data to encode the resolvent trace, the finite-dimensional determinant in the zeta comparison, and the geometry of the Weyl curve developed below.

The Grassmannian geometry of closed cone-operator extensions, including the holomorphic family obtained from null spaces modulo the minimal domain, was developed by Gil, Krainer, and Mendoza [9]. The geometric construction used here arose from organizing the preceding conic analysis in intrinsic boundary coordinates. After completing this construction, we became aware of Wang's abstract Weyl curve for a symmetric operator [22]. The points of contact include the Grassmannian-valued boundary data of formal solutions, Weyl functions as local coordinates, and the associated characteristic bundles. Whereas Wang emphasizes the abstract classification and complex geometry of symmetric operators, our focus is the explicitly normalized critical asymptotic data of conic Laplacians and their relation to Kreĭn's resolvent formula, the resolvent trace identity, zeta-determinant comparison, and the induced determinant line. The critical boundary data of formal solutions define a holomorphic Weyl curve $\mathcal{K}_X : \Sigma_X \to \mathrm{Gr}_m(V_S)$. The Weyl functions are its local graph coordinates. On the real locus, $\mathcal{K}_X$ is a positive curve in $\mathrm{LGr}(V_S, \omega_S)$, whose tangent form is identified with the $L^2(X')$-inner product through the Poisson operator. Furthermore, we identify $d^X_{\mathcal{V},\mathcal{W}}$ with a scalar transition function of the determinant line of the pullback tautological bundle $\mathcal{E}_X$, and show that the corrected zeta determinants form the local representatives of a smooth section of $\mathrm{Det}(\mathcal{E}^{\mathbb{R}}_X) \otimes \mathcal{L}_C$.

The explicit construction of the critical boundary space is two-dimensional. Once this space and its Green form are identified, however, the subsequent Weyl-function,

resolvent, and determinant arguments depend only on finite-dimensional boundary symplectic data and extend to analogous settings subject to the stated spectral and zeta-regularity assumptions.

The paper is organized as follows. Section 2 constructs the critical asymptotic boundary space and the associated Green form for conic Laplacians on compact Riemann surfaces. Section 3 describes self-adjoint realizations in terms of Lagrangian boundary conditions and introduces the corresponding Poisson operators and Weyl functions. Section 4 proves Kreĭn's resolvent formula and the associated trace identity. Section 5 establishes the comparison formula for positive-spectrum zeta determinants. Section 6 introduces the Weyl curve, proves the positivity of its real restriction, and interprets the boundary determinants and corrected zeta determinants in terms of the associated determinant line bundle.

2. Conic Laplacians and Boundary Symplectic Spaces

We construct the boundary symplectic space of a conic Laplacian on a compact Riemann surface. Although the underlying domain theory is standard for regular-singular elliptic operators, a direct Fourier-mode description fixes the notation and normalization needed below. We use the general cone-domain theory only to identify the localized remainder and establish uniqueness; the critical modes and the normalization of their Green pairing are computed explicitly.

Let X be a compact Riemann surface, let $S \subset X$ be a finite set, and set $X' := X \setminus S$. A conic metric on X with singular set S is a Hermitian metric ds_X^2 on X' with the following property: for every $P \in S$, there exist a neighborhood U_P of P, a holomorphic coordinate $z : U_P \to \mathbb{C}$ centered at P, an integer $n_P \geq 2$, and a smooth positive function $\rho_P \in C^\infty(U_P)$ such that

$$ds_X^2 = \rho_P(z)|z|^{2n_P-2}|dz|^2$$

on $U_P' := U_P \setminus \{P\}$. The integer n_P is called the index at P, and the corresponding cone angle is $2\pi n_P$. The points of S are called the conic points of X.

We further assume that ds_X^2 is radial near each conic point. More precisely, for every $P \in S$, there exists a coordinate disk $D_P \subset U_P$ centered at P such that the conformal factor $\rho_P(z)$ depends only on $r = |z|$ on $D_P' := D_P \setminus \{P\}$. Thus, we may write $\rho_P(z) = \rho_P(r)$ on D_P'. Throughout this section, all conic metrics are assumed to be radial in this sense.

Let dA_X denote the area form associated with ds_X^2. The nonnegative Laplace–Beltrami operator, initially defined on compactly supported smooth functions, is denoted by

$$\Delta_{X,c} : C_c^\infty(X') \subset L^2(X', dA_X) \longrightarrow L^2(X', dA_X).$$

With respect to a local coordinate z centered at a conic point P, it is given by

$$\Delta_{X,c} = -\frac{4}{\rho_P(z)|z|^{2n_P-2}} \frac{\partial^2}{\partial z\, \partial \overline{z}}.$$

Writing $z = re^{i\theta}$ on D_P', the radiality assumption gives

$$\Delta_{X,c} = -\frac{1}{\rho_P(r) r^{2n_P}} \left(r^2\partial_r^2 + r\partial_r + \partial_\theta^2 \right).$$

Here $C_c^\infty(X')$ denotes the space of compactly supported smooth complex-valued functions on X'.

Let $L^2(X', dA_X)$ denote the Hilbert space of equivalence classes of measurable complex-valued functions u on X' satisfying $\|u\|^2_{L^2(X',dA_X)} := \int_{X'} |u|^2\, dA_X < \infty$, where functions that agree almost everywhere are identified. Here dA_X denotes the area form associated with the metric ds^2_X. We equip $L^2(X', dA_X)$ with the Hermitian inner product $\langle u, v\rangle_{L^2(X',dA_X)} := \int_{X'} u\overline{v}\, dA_X$. For simplicity, we write $L^2(X') := L^2(X', dA_X)$.

Let $u \in L^2(X')$. We say that the distributional Laplacian of u belongs to $L^2(X')$ if there exists $F \in L^2(X')$ such that $\langle u, \Delta_{X,c}\phi\rangle_{L^2(X')} = \langle F, \phi\rangle_{L^2(X')}$ for every $\phi \in C_c^\infty(X')$. In this case, F is unique and is denoted by $\Delta_X u$.

We now define the maximal and minimal domains of the Laplacian. The maximal domain is $D_{\max}(\Delta_X) := \{u \in L^2(X') : \Delta_X u \in L^2(X')\}$, where $\Delta_X u$ is understood in the distributional sense. Equipped with the graph norm $\|u\|_{\Delta_X} := \big(\|u\|^2_{L^2(X')} + \|\Delta_X u\|^2_{L^2(X')}\big)^{1/2}$, the space $D_{\max}(\Delta_X)$ is a Hilbert space. The minimal domain $D_{\min}(\Delta_X)$ is the closure of $C_c^\infty(X')$ in $D_{\max}(\Delta_X)$ with respect to the graph norm $\|\cdot\|_{\Delta_X}$. Consequently, it is a closed Hilbert subspace of $D_{\max}(\Delta_X)$.

We shall identify the finite-dimensional quotient $D_{\max}(\Delta_X)/D_{\min}(\Delta_X)$ with the critical asymptotic boundary data at the conic points. This will also show that $D_{\min}(\Delta_X) \subsetneq D_{\max}(\Delta_X)$ whenever $S \neq \varnothing$.

Lemma 2.1. *Let $w \in D_{\max}(\Delta_X)$. If $\operatorname{supp} w \Subset X'$, then $w \in D_{\min}(\Delta_X)$.*

Proof. Set $K := \operatorname{supp} w$, and choose relatively compact open sets $\Omega_1, \Omega_2 \subset X'$ such that $K \Subset \Omega_1 \Subset \Omega_2 \Subset X'$. Since $w \in D_{\max}(\Delta_X)$, both w and $\Delta_X w$ belong to $L^2(\Omega_2)$. The interior elliptic estimate gives $\|w\|_{H^2(\Omega_1)} \le C\big(\|\Delta_X w\|_{L^2(\Omega_2)} + \|w\|_{L^2(\Omega_2)}\big)$, and hence $w \in H^2(\Omega_1)$.

Since $\operatorname{supp} w \Subset \Omega_1$, standard smoothing yields a sequence $w_j \in C_c^\infty(\Omega_1)$ such that $w_j \to w$ in $H^2(\Omega_1)$. Since Δ_X has smooth coefficients on Ω_1, it follows that $\Delta_{X,c} w_j \to \Delta_X w$ in $L^2(\Omega_1)$. Moreover, differential operators do not enlarge supports, so w_j, w, $\Delta_{X,c} w_j$, and $\Delta_X w$ all vanish on $X' \setminus \Omega_1$. Consequently, $w_j \to w$ and $\Delta_{X,c} w_j \to \Delta_X w$ in $L^2(X')$. Thus $w_j \to w$ with respect to the graph norm $\|\cdot\|_{\Delta_X}$, and therefore $w \in D_{\min}(\Delta_X)$. □

Lemma 2.2. *Fix $P \in S$, and let D'_P be a punctured coordinate disk centered at P on which $ds^2_X = \rho_P(r) r^{2n_P-2}|dz|^2$, where $z = re^{i\theta}$. Thus $dA_X = \rho_P(r) r^{2n_P-1}\, dr\, d\theta$.*

For $u \in L^2(X')$, write its restriction to D'_P as

$$u(r,\theta) = \sum_{j\in\mathbb{Z}} u_j(r) e^{ij\theta},$$

where

$$u_j(r) := \frac{1}{2\pi}\int_0^{2\pi} u(r,\theta) e^{-ij\theta}\, d\theta.$$

Then, on D'_P,

$$\Delta_X u = -\sum_{j\in\mathbb{Z}} \frac{1}{\rho_P(r) r^{2n_P}} \left(r^2 u_j''(r) + r u_j'(r) - j^2 u_j(r)\right) e^{ij\theta}$$

in the sense of distributions, where the derivatives of u_j are understood distributionally.

The assertion follows directly from the standard Fourier decomposition of the Laplace–Beltrami operator in polar coordinates.

Fix $P \in S$, and abbreviate $D' := D'_P$, $n := n_P$, and $\rho := \rho_P$. Let $F \in L^2(D', dA_X)$, and suppose that $\Delta_X u = F$ in the distributional sense for some $u \in L^2(D', dA_X)$. Write $F(r,\theta) = \sum_{j\in\mathbb{Z}} F_j(r) e^{ij\theta}$ and $u(r,\theta) = \sum_{j\in\mathbb{Z}} u_j(r) e^{ij\theta}$. Then, for every $j \in \mathbb{Z}$,

$$r^2 u_j''(r) + r u_j'(r) - j^2 u_j(r) = -\rho(r) r^{2n} F_j(r) \tag{2.1}$$

in the sense of distributions on $(0,1)$.

Set $f_j(r) := \rho(r) r^{2n-1} F_j(r)$. Since $F_j \in L^2((0,1), \rho(r) r^{2n-1}\, dr)$, the Cauchy–Schwarz inequality gives $f_j \in L^1(0,1)$. Moreover, $f_j \in L^2_{\mathrm{loc}}(0,1)$. It follows from the standard one-dimensional interior regularity that $u_j \in H^2_{\mathrm{loc}}(0,1)$, and (2.1) holds almost everywhere.

Fix $r_0 \in (0,1)$. For $0 < r < r_0$, the zero mode has the form

$$u_0(r) = a_0 + b_0 \log r - \int_0^r \log \frac{r}{\tau}\, f_0(\tau)\, d\tau.$$

For $j \neq 0$, one initially obtains

$$\begin{aligned} u_j(r) = {} & a_j r^{|j|} + b_j r^{-|j|} \\ & + \frac{1}{2|j|} \left(r^{-|j|} \int_0^r \tau^{|j|} f_j(\tau)\, d\tau + r^{|j|} \int_r^{r_0} \tau^{-|j|} f_j(\tau)\, d\tau \right), \end{aligned}$$

where $a_j, b_j \in \mathbb{C}$.

Set $d\mu(r) := \rho(r) r^{2n-1}\, dr$. Weighted Cauchy–Schwarz estimates show that the integral terms belong to $L^2((0,r_0), d\mu)$. Moreover, $r^{-|j|}$ belongs to this space if and only if $|j| < n$. Since $u \in L^2(D', dA_X)$, it follows that $b_j = 0$ whenever $|j| \geq n$.

If $1 \leq |j| \leq n-1$, another weighted Cauchy–Schwarz estimate shows that the function $\tau \mapsto \tau^{-|j|} f_j(\tau)$ belongs to $L^1(0,r_0)$. Absorbing the resulting multiple of $r^{|j|}$ into $a_j r^{|j|}$, we may therefore write

$$\begin{aligned} u_j(r) = {} & a_j r^{|j|} + b_j r^{-|j|} \\ & + \frac{1}{2|j|} \left(r^{-|j|} \int_0^r \tau^{|j|} f_j(\tau)\, d\tau - r^{|j|} \int_0^r \tau^{-|j|} f_j(\tau)\, d\tau \right). \end{aligned}$$

This normalization separates the two homogeneous critical terms from the particular solution.

We now return to the study of $D_{\max}(\Delta_X)$. We first consider the case in which $S = \{P\}$ consists of a single conic point. Fix a coordinate disk $D_P \Subset U_P$ centered at P on which the metric is radial, and write $z = re^{i\theta}$.

Theorem 2.3. *Assume that $S = \{P\}$. For every $u \in D_{\max}(\Delta_X)$, there exist unique coefficients $a_j, b_j \in \mathbb{C}$, $|j| \leq n_P - 1$, such that, on $D'_P := D_P \setminus \{P\}$,*

$$u(r,\theta) = a_0 + b_0 \log r + \sum_{1 \leq |j| \leq n_P - 1} \left(a_j r^{|j|} + b_j r^{-|j|} \right) e^{ij\theta} + R(r,\theta).$$

Moreover, for every cutoff function $\chi \in C_c^\infty(D_P)$ equal to 1 in a neighborhood of P, the function χR, extended by zero to X', belongs to $D_{\min}(\Delta_X)$.

Proof. Set $F := \Delta_X u \in L^2(X')$. After shrinking D_P, if necessary, we identify it with the unit disk by the coordinate $z = re^{i\theta}$. On D_P', write $u(r,\theta) = \sum_{j\in\mathbb{Z}} u_j(r)e^{ij\theta}$ and $F(r,\theta) = \sum_{j\in\mathbb{Z}} F_j(r)e^{ij\theta}$.

The preceding modewise analysis shows that there exist coefficients $a_j, b_j \in \mathbb{C}$, $|j| \le n_P - 1$, such that

$$u(r,\theta) = a_0 + b_0 \log r + \sum_{1\le|j|\le n_P-1} \left(a_j r^{|j|} + b_j r^{-|j|}\right) e^{ij\theta} + R(r,\theta).$$

Indeed, the L^2-condition implies that the coefficient of $r^{-|j|}e^{ij\theta}$ vanishes whenever $|j| \ge n_P$.

It remains to identify the localized remainder with an element of the minimal domain. Near P, the Laplacian is the b-elliptic cone operator

$$-r^{-2n_P}\rho_P(r)^{-1}\left((r\partial_r)^2 + \partial_\theta^2\right)$$

acting in $L^2(\rho_P(r)r^{2n_P}\,dr/r\,d\theta)$. On the angular mode $e^{ij\theta}$, the indicial polynomial of the factor in parentheses is $\zeta^2 - j^2$. In the exponent convention r^ζ, the critical strip is $-n_P < \operatorname{Re}\zeta < n_P$. Thus its indicial terms are exactly

$$1, \quad \log r, \quad r^{|j|}e^{ij\theta}, \quad r^{-|j|}e^{ij\theta}, \qquad 1 \le |j| \le n_P - 1.$$

The modewise formulas above show that these are precisely the critical Mellin singular parts of u. To state the resulting estimate for the remainder precisely, let H_b^2 denote the local b-Sobolev space defined using the vector fields $r\partial_r$ and ∂_θ. In the notation of [8], the order is $m = 2$ and the weight of the present cone operator is $\nu = 2n_P$. After localization near P, Proposition 3.6 of [8] says that an element $w \in D_{\max}(\Delta_X)$ supported in D_P, and extended by zero to X', belongs to the minimal domain precisely when

$$w \in \bigcap_{\varepsilon>0} r^{n_P-\varepsilon}H_b^2.$$

More precisely, Lemma 7.6 and the discussion preceding it in [8] show that removing all the critical Mellin singular parts leaves a remainder satisfying

$$\chi R \in r^{n_P-\varepsilon}H_b^2 \qquad \text{for every } \varepsilon > 0.$$

Here this is a weighted b-Sobolev statement, rather than merely a pointwise estimate. Moreover, $\chi R \in D_{\max}(\Delta_X)$, since it is the difference between χu and the finitely many cutoff critical terms displayed above. Proposition 3.6 therefore gives $\chi R \in D_{\min}(\Delta_X)$. The same identification shows that the critical coefficients are unique. □

Motivated by the preceding asymptotic expansion, we introduce the following definition.

Definition 2.4. Let $P \in S$ be a conic point with index n_P, and let z_P be the chosen local coordinate centered at P. Write $z_P = r_P e^{i\theta_P}$, where $\theta_P \in \mathbb{R}/2\pi\mathbb{Z}$. Define $f_{P,0} := 1/\sqrt{2\pi}$ and $f_{P,0}^{\#} := -\log r_P/\sqrt{2\pi}$. For $1 \le |j| \le n_P - 1$, define $f_{P,j} := r_P^{|j|}e^{ij\theta_P}/\sqrt{4\pi|j|}$ and $f_{P,j}^{\#} := r_P^{-|j|}e^{ij\theta_P}/\sqrt{4\pi|j|}$.

The critical asymptotic space at P, with respect to z_P, is

$$V_P := \operatorname{span}_{\mathbb{C}}\{f_{P,j}, f^{\#}_{P,j} : |j| \leq n_P - 1\}.$$

The elements $f_{P,j}$ and $f^{\#}_{P,j}$, $|j| \leq n_P - 1$, are called the critical asymptotic terms at P.

Corollary 2.5. *Assume that $S = \{P\}$, and let D_P be the coordinate disk fixed above. For every $u \in D_{\max}(\Delta_X)$, there exists a unique element $h \in V_P$ such that, for every cutoff function $\chi \in C^\infty_c(D_P)$ equal to 1 near P, the function $R_\chi := u - \chi h$ belongs to $D_{\min}(\Delta_X)$. Here χh is extended by zero to X'. Consequently, $u = \chi h + R_\chi$, where $h \in V_P$ and $R_\chi \in D_{\min}(\Delta_X)$.*

Proof. By Theorem 2.3, there exists a unique $h \in V_P$ such that $u = h + R_0$ on D'_P, where $\chi R_0 \in D_{\min}(\Delta_X)$ for every cutoff function χ as above. For any such χ, one has $R_\chi = u - \chi h = \chi R_0 + (1 - \chi)u$.

The first term belongs to $D_{\min}(\Delta_X)$ by Theorem 2.3. By interior elliptic regularity away from P, the second term belongs to $D_{\max}(\Delta_X)$. Moreover, it vanishes near P, and hence its support is compactly contained in X'. Lemma 2.1 therefore gives $(1 - \chi)u \in D_{\min}(\Delta_X)$. It follows that $R_\chi \in D_{\min}(\Delta_X)$. □

The unique element $h \in V_P$ appearing in the preceding corollary is called the critical asymptotic part of u at P.

Corollary 2.6. *Assume that $S = \{P\}$. The subspace $C^\infty(X') \cap D_{\max}(\Delta_X)$ is dense in $D_{\max}(\Delta_X)$ with respect to the graph norm.*

Proof. Let $u \in D_{\max}(\Delta_X)$, and let $h \in V_P$ be its critical asymptotic part at P. Choose a cutoff function $\chi \in C^\infty_c(D_P)$ equal to 1 near P, and set $R := u - \chi h$, where χh is extended by zero outside D_P. By Corollary 2.5, one has $R \in D_{\min}(\Delta_X)$.

Since $D_{\min}(\Delta_X)$ is the graph-norm closure of $C^\infty_c(X')$, there exists a sequence $R_j \in C^\infty_c(X')$ such that $\|R_j - R\|_{\Delta_X} \to 0$. Set $u_j := \chi h + R_j$. Since $\chi h = u - R \in D_{\max}(\Delta_X)$ and χh is smooth on X', we have $u_j \in C^\infty(X') \cap D_{\max}(\Delta_X)$. Moreover, $\|u_j - u\|_{\Delta_X} = \|R_j - R\|_{\Delta_X} \to 0$. This proves the result. □

We now introduce the Green form on $D_{\max}(\Delta_X)$. For $u, v \in D_{\max}(\Delta_X)$, define

$$q_X(u, v) := \langle \Delta_X u, v\rangle_{L^2(X')} - \langle u, \Delta_X v\rangle_{L^2(X')}.$$

Then q_X is skew-Hermitian; that is, $q_X(v, u) = -\overline{q_X(u, v)}$.

Lemma 2.7. *For every $u \in D_{\max}(\Delta_X)$ and $v \in D_{\min}(\Delta_X)$, one has $q_X(u, v) = q_X(v, u) = 0$. Consequently, q_X induces a well-defined skew-Hermitian sesquilinear form on $D_{\max}(\Delta_X)/D_{\min}(\Delta_X)$.*

Proof. Let $v \in D_{\min}(\Delta_X)$. By the definition of the minimal domain, there exists a sequence $v_j \in C^\infty_c(X')$ such that $v_j \to v$ and $\Delta_{X,c} v_j \to \Delta_X v$ in $L^2(X')$. For every $u \in D_{\max}(\Delta_X)$, the definition of the distributional Laplacian gives $\langle \Delta_X u, v_j\rangle_{L^2(X')} = \langle u, \Delta_{X,c} v_j\rangle_{L^2(X')}$. Thus $q_X(u, v_j) = 0$. Passing to the limit yields $q_X(u, v) = 0$. The skew-Hermitian property then gives $q_X(v, u) = 0$.

Therefore, q_X is unchanged when either argument is modified by an element of $D_{\min}(\Delta_X)$, and hence it induces a well-defined form on $D_{\max}(\Delta_X)/D_{\min}(\Delta_X)$. □

We define the local boundary form $\omega_P : V_P \times V_P \to \mathbb{C}$ by requiring $\omega_P(f_{P,j}, f_{P,k}^{\#}) = \delta_{jk}$, $\omega_P(f_{P,j}^{\#}, f_{P,k}) = -\delta_{jk}$, and $\omega_P(f_{P,j}, f_{P,k}) = \omega_P(f_{P,j}^{\#}, f_{P,k}^{\#}) = 0$ for $|j|, |k| \leq n_P - 1$.

Equivalently, if $h = \sum_{|j|\leq n_P-1}(a_j f_{P,j} + b_j f_{P,j}^{\#})$ and $\ell = \sum_{|j|\leq n_P-1}(c_j f_{P,j} + d_j f_{P,j}^{\#})$, then

$$\omega_P(h, \ell) = \sum_{|j|\leq n_P-1} \left(a_j \overline{d_j} - b_j \overline{c_j}\right).$$

Lemma 2.8. *The sesquilinear form ω_P on V_P is nondegenerate and skew-Hermitian.*

Proof. Let $h = \sum_{|j|\leq n_P-1}(a_j f_{P,j} + b_j f_{P,j}^{\#})$ and $\ell = \sum_{|j|\leq n_P-1}(c_j f_{P,j} + d_j f_{P,j}^{\#})$. By definition, $\omega_P(h, \ell) = \sum_{|j|\leq n_P-1}(a_j \overline{d_j} - b_j \overline{c_j})$. It follows immediately that $\omega_P(\ell, h) = -\overline{\omega_P(h, \ell)}$, so ω_P is skew-Hermitian.

Suppose that $\omega_P(h, \ell) = 0$ for every $\ell \in V_P$. Taking $\ell = f_{P,j}^{\#}$ gives $a_j = 0$, while taking $\ell = f_{P,j}$ gives $b_j = 0$, for every $|j| \leq n_P - 1$. Thus $h = 0$, and hence ω_P is nondegenerate. □

Lemma 2.9. *Assume that $S = \{P\}$. Let $u, v \in D_{\max}(\Delta_X)$, and let $h_u, h_v \in V_P$ be their critical asymptotic parts. Then $q_X(u, v) = \omega_P(h_u, h_v)$.*

Proof. Choose a cutoff function $\chi \in C_c^\infty(D_P)$ equal to 1 near P. By Corollary 2.5, write $u = \chi h_u + R_u$ and $v = \chi h_v + R_v$, where $R_u, R_v \in D_{\min}(\Delta_X)$. Lemma 2.7 gives $q_X(u, v) = q_X(\chi h_u, \chi h_v)$. A direct application of Green's formula to the normalized critical asymptotic terms gives $q_X(\chi f_{P,j}, \chi f_{P,k}^{\#}) = \delta_{jk}$, $q_X(\chi f_{P,j}^{\#}, \chi f_{P,k}) = -\delta_{jk}$, and $q_X(\chi f_{P,j}, \chi f_{P,k}) = q_X(\chi f_{P,j}^{\#}, \chi f_{P,k}^{\#}) = 0$ for $|j|, |k| \leq n_P - 1$. Comparing these identities with the definition of ω_P and using sesquilinearity yields $q_X(u, v) = \omega_P(h_u, h_v)$. □

Corollary 2.10. *Assume that $S = \{P\}$. Let $\chi \in C_c^\infty(D_P)$ be equal to 1 near P. Then $\chi V_P \cap D_{\min}(\Delta_X) = \{0\}$. Equivalently, if $h \in V_P$ is nonzero, then $\chi h \notin D_{\min}(\Delta_X)$. In particular, $\chi f_{P,j}$ and $\chi f_{P,j}^{\#}$ do not belong to $D_{\min}(\Delta_X)$ for any $|j| \leq n_P - 1$.*

Proof. Suppose that $h \in V_P$ and $\chi h \in D_{\min}(\Delta_X)$. For every $\ell \in V_P$, the function $\chi \ell$ belongs to $D_{\max}(\Delta_X)$. Hence Lemma 2.7 gives $q_X(\chi h, \chi \ell) = 0$. On the other hand, Lemma 2.9 gives $q_X(\chi h, \chi \ell) = \omega_P(h, \ell)$. Thus $\omega_P(h, \ell) = 0$ for every $\ell \in V_P$. Since ω_P is nondegenerate, $h = 0$. □

Corollary 2.11. *Assume that $S = \{P\}$. Then $D_{\min}(\Delta_X) \subsetneq D_{\max}(\Delta_X)$.*

Proof. Choose a cutoff function $\chi \in C_c^\infty(D_P)$ equal to 1 near P. Then $\chi f_{P,0} \in D_{\max}(\Delta_X)$, whereas Corollary 2.10 gives $\chi f_{P,0} \notin D_{\min}(\Delta_X)$. Hence the inclusion is strict. □

The left radical of q_X is defined by $\operatorname{rad}_L q_X := \{u \in D_{\max}(\Delta_X) : q_X(u, v) = 0 \text{ for every } v \in D_{\max}(\Delta_X)\}$. Similarly, its right radical is $\operatorname{rad}_R q_X := \{v \in D_{\max}(\Delta_X) : q_X(u, v) = 0 \text{ for every } u \in D_{\max}(\Delta_X)\}$. Since q_X is skew-Hermitian, these two subspaces coincide. We denote their common value by $\operatorname{rad} q_X$. By Lemma 2.7, one has $D_{\min}(\Delta_X) \subset \operatorname{rad} q_X$.

Corollary 2.12. *The radical of q_X is precisely the minimal domain:* $\operatorname{rad} q_X = D_{\min}(\Delta_X)$.

Proof. Lemma 2.7 gives $D_{\min}(\Delta_X) \subset \operatorname{rad} q_X$. Conversely, let $u \in \operatorname{rad} q_X$, and let $h_u \in V_P$ be its critical asymptotic part at P. Choose $\chi \in C_c^\infty(D_P)$ equal to 1 near P, and set $R := u - \chi h_u$. By Corollary 2.5, one has $R \in D_{\min}(\Delta_X)$. For every $\ell \in V_P$, the function $\chi\ell$ belongs to $D_{\max}(\Delta_X)$. Since $u \in \operatorname{rad} q_X$, one has $q_X(u, \chi\ell) = 0$. Lemma 2.7 and Lemma 2.9 give

$$0 = q_X(u, \chi\ell) = q_X(\chi h_u, \chi\ell) = \omega_P(h_u, \ell).$$

Since ω_P is nondegenerate, $h_u = 0$. Therefore $u = R \in D_{\min}(\Delta_X)$, proving the reverse inclusion. □

Theorem 2.13. *Assume that $S = \{P\}$. Let $\pi_P : D_{\max}(\Delta_X) \to V_P$ be the map that assigns to each $u \in D_{\max}(\Delta_X)$ its critical asymptotic part at P. Then π_P is a surjective linear map and* $\ker \pi_P = D_{\min}(\Delta_X)$. *Consequently, π_P induces a linear isomorphism*

$$\overline{\pi}_P : D_{\max}(\Delta_X)/D_{\min}(\Delta_X) \longrightarrow V_P.$$

In particular, $\dim_{\mathbb{C}}\big(D_{\max}(\Delta_X)/D_{\min}(\Delta_X)\big) = 4n_P - 2$.

Proof. The linearity of π_P follows from the uniqueness of the critical asymptotic part. We first identify the kernel of π_P. By Corollary 2.5, an element $u \in D_{\max}(\Delta_X)$ has vanishing critical asymptotic part if and only if $u \in D_{\min}(\Delta_X)$. Therefore, $\ker \pi_P = D_{\min}(\Delta_X)$. To prove surjectivity, let $h \in V_P$, and choose $\chi \in C_c^\infty(D_P)$ equal to 1 near P. Then $\chi h \in D_{\max}(\Delta_X)$, and its critical asymptotic part is h. Hence $\pi_P(\chi h) = h$. The asserted isomorphism now follows from the first isomorphism theorem. Finally, $\dim_{\mathbb{C}} V_P = 2(2n_P - 1) = 4n_P - 2$. □

We now return to the case of an arbitrary finite conic set S.

Lemma 2.14. *Let $\chi_P \in C_c^\infty(D_P)$, $P \in S$, be cutoff functions equal to 1 near P, with pairwise disjoint supports. For $u \in D_{\max}(\Delta_X)$, let $h_{u,P} \in V_P$ be its critical asymptotic part at P. Then*

$$R := u - \sum_{P \in S} \chi_P h_{u,P} \in D_{\min}(\Delta_X).$$

Proof. Set $\eta := 1 - \sum_{P \in S} \chi_P$. Since the supports of the cutoff functions are pairwise disjoint and each χ_P is equal to 1 near P, the function η vanishes in a neighborhood of S. Since X is compact, it follows that $\operatorname{supp}\eta \Subset X'$, and hence $\operatorname{supp}(\eta u) \Subset X'$. By interior elliptic regularity, $u \in H^2_{\mathrm{loc}}(X')$. Therefore, $\eta u \in D_{\max}(\Delta_X)$. Lemma 2.1 then gives $\eta u \in D_{\min}(\Delta_X)$. Moreover,

$$R = u - \sum_{P \in S} \chi_P h_{u,P} = \sum_{P \in S} \chi_P (u - h_{u,P}) + \eta u.$$

By the local remainder statement, $\chi_P(u - h_{u,P}) \in D_{\min}(\Delta_X)$ for every $P \in S$. Since S is finite, it follows that $R \in D_{\min}(\Delta_X)$. □

For each $P \in S$, let V_P and ω_P be the critical asymptotic space and the normalized local Green form defined above. Set $V_S := \bigoplus_{P\in S} V_P$. For $h = (h_P)_{P\in S}$ and $\ell = (\ell_P)_{P\in S}$ in V_S, define $\omega_S(h,\ell) := \sum_{P\in S} \omega_P(h_P, \ell_P)$. Then ω_S is a nondegenerate skew-Hermitian sesquilinear form on V_S.

For each $P \in S$, let $\pi_P : D_{\max}(\Delta_X) \to V_P$ be the map that assigns to $u \in D_{\max}(\Delta_X)$ its critical asymptotic part at P. Define the global critical asymptotic map $\pi_S : D_{\max}(\Delta_X) \to V_S$ by $\pi_S(u) := (\pi_P(u))_{P\in S}$.

Lemma 2.15. *For every $u, v \in D_{\max}(\Delta_X)$, one has $q_X(u,v) = \omega_S\big(\pi_S(u), \pi_S(v)\big)$.*

Proof. Choose cutoff functions $\chi_P \in C_c^\infty(D_P)$, $P \in S$, equal to 1 near P, with pairwise disjoint supports. By Lemma 2.14, write

$$u = \sum_{P\in S} \chi_P h_{u,P} + R_u, \qquad v = \sum_{P\in S} \chi_P h_{v,P} + R_v,$$

where $R_u, R_v \in D_{\min}(\Delta_X)$ and $h_{u,P} = \pi_P(u)$, $h_{v,P} = \pi_P(v)$. Lemma 2.7 gives

$$q_X(u,v) = q_X\left(\sum_{P\in S} \chi_P h_{u,P}, \sum_{P\in S} \chi_P h_{v,P}\right).$$

Since the supports of the cutoff functions are pairwise disjoint, all cross terms corresponding to distinct conic points vanish. Hence

$$q_X(u,v) = \sum_{P\in S} q_X(\chi_P h_{u,P}, \chi_P h_{v,P}).$$

By the local Green formula, $q_X(\chi_P h_{u,P}, \chi_P h_{v,P}) = \omega_P(h_{u,P}, h_{v,P})$. Therefore,

$$q_X(u,v) = \sum_{P\in S} \omega_P\big(\pi_P(u), \pi_P(v)\big) = \omega_S\big(\pi_S(u), \pi_S(v)\big).$$

□

Corollary 2.16. *For each $P \in S$, let $\chi_P \in C_c^\infty(D_P)$ be equal to 1 near P. Then, for every $u \in D_{\max}(\Delta_X)$,*

$$\pi_P(u) = \sum_{|j|\le n_P - 1} \Big(q_X(u, \chi_P f_{P,j}^{\#}) f_{P,j} - q_X(u, \chi_P f_{P,j}) f_{P,j}^{\#} \Big).$$

In particular, the right-hand side is independent of the choice of χ_P.

Proof. Write

$$\pi_P(u) = \sum_{|j|\le n_P - 1} \Big(a_j f_{P,j} + b_j f_{P,j}^{\#} \Big).$$

By Lemma 2.15 and the normalization of the critical asymptotic terms, one has

$$q_X(u, \chi_P f_{P,j}^{\#}) = a_j, \qquad q_X(u, \chi_P f_{P,j}) = -b_j.$$

Substituting these identities into the expansion of $\pi_P(u)$ gives the asserted formula. Since χ_P was arbitrary, the right-hand side is independent of its choice. □

Theorem 2.17. *The map π_S is a surjective linear map with $\ker \pi_S = D_{\min}(\Delta_X)$. Consequently, it induces a linear isomorphism*

$$\overline{\pi}_S : D_{\max}(\Delta_X)/D_{\min}(\Delta_X) \longrightarrow V_S, \qquad [u] \longmapsto \pi_S(u).$$

In particular, $\dim_{\mathbb{C}}\big(D_{\max}(\Delta_X)/D_{\min}(\Delta_X)\big) = \sum_{P\in S}(4n_P - 2)$.

Proof. The linearity of π_S follows immediately from Corollary 2.16.

We first prove surjectivity. Choose coordinate disks D_P, $P \in S$, with pairwise disjoint closures, and cutoff functions $\chi_P \in C_c^\infty(D_P)$ equal to 1 near P. Let $h = (h_P)_{P\in S} \in V_S$, and define $u := \sum_{P\in S} \chi_P h_P$, where each $\chi_P h_P$ is extended by zero outside D_P. Then $u \in D_{\max}(\Delta_X)$ and $\pi_P(u) = h_P$ for every $P \in S$. Hence $\pi_S(u) = h$, proving that π_S is surjective.

We next determine the kernel. Suppose first that $u \in \ker \pi_S$. Then $\pi_P(u) = 0$ for every $P \in S$. Lemma 2.14 gives

$$u = u - \sum_{P\in S} \chi_P \pi_P(u) \in D_{\min}(\Delta_X).$$

Thus $\ker \pi_S \subset D_{\min}(\Delta_X)$. Conversely, let $u \in D_{\min}(\Delta_X)$. For every $v \in D_{\max}(\Delta_X)$, Lemma 2.7 and Lemma 2.15 give

$$0 = q_X(u, v) = \omega_S\big(\pi_S(u), \pi_S(v)\big).$$

Since π_S is surjective, $\pi_S(v)$ ranges over all of V_S. The nondegeneracy of ω_S therefore implies $\pi_S(u) = 0$. Hence $u \in \ker \pi_S$, proving that $\ker \pi_S = D_{\min}(\Delta_X)$.

The asserted isomorphism follows from the first isomorphism theorem. Finally,

$$\dim_{\mathbb{C}} V_S = \sum_{P\in S} \dim_{\mathbb{C}} V_P = \sum_{P\in S}(4n_P - 2).$$

□

3. Self-Adjoint Extensions and the Weyl Function

We use (V_S, ω_S) to parametrize the self-adjoint realizations by Lagrangian subspaces and to construct their Poisson operators and Weyl functions.

Definition 3.1. For a subspace $\mathcal{V} \subset V_S$, define its ω_S-orthogonal complement by

$$\mathcal{V}^{\perp_{\omega_S}} := \{h \in V_S : \omega_S(h, \ell) = 0 \text{ for every } \ell \in \mathcal{V}\}.$$

The subspace $\mathcal{V}$ is called isotropic if $\mathcal{V} \subset \mathcal{V}^{\perp_{\omega_S}}$, and Lagrangian if $\mathcal{V} = \mathcal{V}^{\perp_{\omega_S}}$. We denote by

$$\mathrm{LGr}(V_S, \omega_S) := \big\{\mathcal{V} \subset V_S : \mathcal{V} = \mathcal{V}^{\perp_{\omega_S}}\big\}$$

the Lagrangian Grassmannian associated with the nondegenerate skew-Hermitian space (V_S, ω_S).

We denote the minimal Laplacian by $\Delta_{X,\min} := \Delta_X\big|_{D_{\min}(\Delta_X)}$. For any subspace $\mathcal{V} \subset V_S$, define

$$D_{\mathcal{V}} := \{u \in D_{\max}(\Delta_X) : \pi_S(u) \in \mathcal{V}\}, \qquad \Delta_{X,\mathcal{V}} := \Delta_X\big|_{D_{\mathcal{V}}}.$$

Since $\ker \pi_S = D_{\min}(\Delta_X)$, one has $D_{\{0\}} = D_{\min}(\Delta_X)$ and $\Delta_{X,\{0\}} = \Delta_{X,\min}$.

Theorem 3.2. *For every subspace $\mathcal{V} \subset V_S$, one has*

$$\Delta_{X,\mathcal{V}}^* = \Delta_{X,\mathcal{V}^{\perp_{\omega_S}}}.$$

In particular, $\Delta_{X,\mathcal{V}}$ is self-adjoint if and only if $\mathcal{V}$ is Lagrangian. Moreover, the assignment $\mathcal{V} \mapsto \Delta_{X,\mathcal{V}}$ gives a bijection from $\mathrm{LGr}(V_S, \omega_S)$ onto the set of all self-adjoint extensions of $\Delta_{X,\min}$.

Proof. Since $\ker \pi_S = D_{\min}(\Delta_X)$, we have $D_{\min}(\Delta_X) \subset D_{\mathcal{V}}$. In particular, $C_c^\infty(X') \subset D_{\mathcal{V}}$, so $\Delta_{X,\mathcal{V}}$ is densely defined.

Let $u \in D(\Delta_{X,\mathcal{V}}^*)$. Testing the defining identity for the adjoint against functions in $C_c^\infty(X')$ shows that $u \in D_{\max}(\Delta_X)$ and $\Delta_{X,\mathcal{V}}^* u = \Delta_X u$. Therefore,

$$\begin{aligned} u \in D(\Delta_{X,\mathcal{V}}^*) &\iff q_X(u,v) = 0 \quad \text{for every } v \in D_{\mathcal{V}} \\ &\iff \omega_S\big(\pi_S(u), \ell\big) = 0 \quad \text{for every } \ell \in \mathcal{V} \\ &\iff \pi_S(u) \in \mathcal{V}^{\perp_{\omega_S}}. \end{aligned}$$

Here we have used $\pi_S(D_{\mathcal{V}}) = \mathcal{V}$, which follows from the surjectivity of π_S. Thus $D(\Delta_{X,\mathcal{V}}^*) = D_{\mathcal{V}^{\perp_{\omega_S}}}$. Since both operators act by Δ_X, it follows that $\Delta_{X,\mathcal{V}}^* = \Delta_{X,\mathcal{V}^{\perp_{\omega_S}}}$.

Consequently, $\Delta_{X,\mathcal{V}}$ is self-adjoint if and only if $D_{\mathcal{V}} = D_{\mathcal{V}^{\perp_{\omega_S}}}$. Since π_S is surjective, this is equivalent to $\mathcal{V} = \mathcal{V}^{\perp_{\omega_S}}$, namely, to $\mathcal{V}$ being Lagrangian.

Finally, let $\widetilde{\Delta}$ be any self-adjoint extension of the minimal conic Laplacian, and set $\mathcal{D} := D(\widetilde{\Delta})$. Since $\widetilde{\Delta}$ is a self-adjoint extension of the minimal Laplacian, one has $\mathcal{D} \subset D_{\max}(\Delta_X)$ and $\widetilde{\Delta} u = \Delta_X u$ for $u \in \mathcal{D}$. Set $\mathcal{V} := \pi_S(\mathcal{D})$.

We claim that $\mathcal{D} = D_{\mathcal{V}}$. The inclusion $\mathcal{D} \subset D_{\mathcal{V}}$ is immediate. Conversely, if $u \in D_{\mathcal{V}}$, choose $v \in \mathcal{D}$ such that $\pi_S(v) = \pi_S(u)$. Then

$$u - v \in \ker \pi_S = D_{\min}(\Delta_X) \subset \mathcal{D},$$

and hence $u \in \mathcal{D}$. Therefore, $\mathcal{D} = D_{\mathcal{V}}$ and $\widetilde{\Delta} = \Delta_{X,\mathcal{V}}$. Since $\widetilde{\Delta}$ is self-adjoint, the preceding result implies that $\mathcal{V}$ is Lagrangian. The Lagrangian subspace is unique because $\mathcal{V} = \pi_S(\mathcal{D})$, and hence the correspondence is bijective. □

The elements of V_S will be called the critical asymptotic boundary data. For a Lagrangian subspace $\mathcal{V} \subset V_S$, the condition $\pi_S(u) \in \mathcal{V}$ will be called the Lagrangian boundary condition determined by $\mathcal{V}$. The corresponding self-adjoint realization of Δ_X, with domain $D_{\mathcal{V}}$, will be denoted by $\Delta_{X,\mathcal{V}}$.

Throughout the remainder of this paper, for $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$ and $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, we denote the resolvent of $\Delta_{X,\mathcal{V}}$ by $R_{\mathcal{V}}(\lambda) := (\lambda - \Delta_{X,\mathcal{V}})^{-1}$.

Theorem 3.3. *Let $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$, and let $\Delta_{X,\mathcal{V}}$ be the corresponding self-adjoint realization. Then $\Delta_{X,\mathcal{V}}$ is bounded from below and has compact resolvent. Consequently, its spectrum is discrete and consists of real eigenvalues of finite multiplicity that can accumulate only at $+\infty$.*

Proof. The compactness of the resolvent follows from the singular elliptic estimate and the Rellich property for closed extensions of elliptic cone operators. More precisely, the maximal domain embeds continuously into a positively weighted cone Sobolev space, which is compactly embedded in the underlying weighted L^2-space; see [8,

Lemma 3.9 and (3.10)]. Moreover, $\Delta_{X,\min}$ is nonnegative and has finite deficiency indices. By the standard extension theory of semibounded symmetric operators with finite deficiency indices, every self-adjoint extension of $\Delta_{X,\min}$ is bounded from below. □

Let $\{(\lambda_i^{\mathcal{V}}, \phi_i^{\mathcal{V}})\}_{i\geq 0}$ be the eigenpairs of $\Delta_{X,\mathcal{V}}$, with the eigenvalues listed in nondecreasing order and repeated according to multiplicity: $\lambda_0^{\mathcal{V}} \leq \lambda_1^{\mathcal{V}} \leq \cdots$ and $\lambda_i^{\mathcal{V}} \to +\infty$. We choose the eigenfunctions so that $\{\phi_i^{\mathcal{V}}\}_{i\geq 0}$ is a complete orthonormal basis of $L^2(X')$. Thus $\Delta_{X,\mathcal{V}}\phi_i^{\mathcal{V}} = \lambda_i^{\mathcal{V}}\phi_i^{\mathcal{V}}$. For each $i \geq 0$, define the linear functional $q_i^{\mathcal{V}} : D_{\max}(\Delta_X) \to \mathbb{C}$ by $q_i^{\mathcal{V}}(u) := q_X(u, \phi_i^{\mathcal{V}})$.

Lemma 3.4. *Let $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$ and $\lambda \in \rho(\Delta_{X,\mathcal{V}})$. For every $u \in D_{\max}(\Delta_X)$, the series $\sum_{i\geq 0}(\lambda - \lambda_i^{\mathcal{V}})^{-1} q_i^{\mathcal{V}}(u)\phi_i^{\mathcal{V}}$ converges in $L^2(X')$. Define*

$$Y_{\mathcal{V}}(\lambda) : D_{\max}(\Delta_X) \longrightarrow L^2(X')$$

by $Y_{\mathcal{V}}(\lambda)u := \sum_{i\geq 0}(\lambda - \lambda_i^{\mathcal{V}})^{-1} q_i^{\mathcal{V}}(u)\phi_i^{\mathcal{V}}$. Then $Y_{\mathcal{V}}(\lambda)u = u + R_{\mathcal{V}}(\lambda)(\Delta_X - \lambda)u$.

Proof. For brevity, write $\langle \cdot, \cdot \rangle$ for the inner product on $L^2(X')$, and set $f := (\Delta_X - \lambda)u$. Since $u \in D_{\max}(\Delta_X)$, one has $f \in L^2(X')$. For each $i \geq 0$,

$$q_i^{\mathcal{V}}(u) = \langle f, \phi_i^{\mathcal{V}} \rangle + (\lambda - \lambda_i^{\mathcal{V}})\langle u, \phi_i^{\mathcal{V}} \rangle.$$

Consequently,

$$\frac{q_i^{\mathcal{V}}(u)}{\lambda - \lambda_i^{\mathcal{V}}} = \langle u, \phi_i^{\mathcal{V}} \rangle + \frac{\langle f, \phi_i^{\mathcal{V}} \rangle}{\lambda - \lambda_i^{\mathcal{V}}}.$$

The first term on the right is the i-th Fourier coefficient of u with respect to the orthonormal basis $\{\phi_i^{\mathcal{V}}\}_{i\geq 0}$, while the second is the corresponding coefficient in the spectral expansion of $R_{\mathcal{V}}(\lambda)f$. Both series converge in $L^2(X')$. Hence $Y_{\mathcal{V}}(\lambda)u = u + R_{\mathcal{V}}(\lambda)f = u + R_{\mathcal{V}}(\lambda)(\Delta_X - \lambda)u$. □

The identity in Lemma 3.4 shows that $Y_{\mathcal{V}}(\lambda)u \in D_{\max}(\Delta_X)$ for every $u \in D_{\max}(\Delta_X)$. Thus we regard $Y_{\mathcal{V}}(\lambda)$ as a linear operator on $D_{\max}(\Delta_X)$.

Corollary 3.5. *Let $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$ and $\lambda \in \rho(\Delta_{X,\mathcal{V}})$. Then $Y_{\mathcal{V}}(\lambda)$ is the projection of $D_{\max}(\Delta_X)$ onto $\ker(\Delta_X - \lambda)$ along $D_{\mathcal{V}}$. Consequently,*

$$D_{\max}(\Delta_X) = D_{\mathcal{V}} \oplus \ker(\Delta_X - \lambda).$$

Moreover, $\operatorname{rank} Y_{\mathcal{V}}(\lambda) = \frac{1}{2}\dim_{\mathbb{C}} V_S$.

Indeed, the identity in Lemma 3.4 shows that $Y_{\mathcal{V}}(\lambda)$ is the identity on $\ker(\Delta_X - \lambda)$ and that $\ker Y_{\mathcal{V}}(\lambda) = D_{\mathcal{V}}$. The rank formula follows from $D_{\max}(\Delta_X)/D_{\mathcal{V}} \cong V_S/\mathcal{V}$ and $\dim_{\mathbb{C}} \mathcal{V} = \frac{1}{2}\dim_{\mathbb{C}} V_S$.

Definition 3.6. A Lagrangian pair in (V_S, ω_S) is an ordered pair $(\mathcal{V}, \mathcal{V}^{\#})$ of Lagrangian subspaces satisfying $V_S = \mathcal{V} \oplus \mathcal{V}^{\#}$.

Let $(\mathcal{V}, \mathcal{V}^{\#})$ be a Lagrangian pair in (V_S, ω_S). Denote by $P_{\mathcal{V}} : V_S \to \mathcal{V}$ and $P_{\mathcal{V}^{\#}} : V_S \to \mathcal{V}^{\#}$ the corresponding projections. Define $\widetilde{\pi}_{\mathcal{V}} := P_{\mathcal{V}} \circ \pi_S : D_{\max}(\Delta_X) \to \mathcal{V}$ and $\widetilde{\pi}_{\mathcal{V}^{\#}} := P_{\mathcal{V}^{\#}} \circ \pi_S : D_{\max}(\Delta_X) \to \mathcal{V}^{\#}$. Thus $\pi_S(u) = \widetilde{\pi}_{\mathcal{V}}(u) + \widetilde{\pi}_{\mathcal{V}^{\#}}(u)$ for every $u \in D_{\max}(\Delta_X)$.

Fix $\lambda \in \rho(\Delta_{X,\mathcal{V}})$. By Corollary 3.5, $Y_{\mathcal{V}}(\lambda)$ is a projection satisfying

$$\ker Y_{\mathcal{V}}(\lambda) = D_{\mathcal{V}}, \qquad \operatorname{ran} Y_{\mathcal{V}}(\lambda) = \ker(\Delta_X - \lambda).$$

Therefore, it induces a linear isomorphism

$$\overline{Y}_{\mathcal{V}}(\lambda) : D_{\max}(\Delta_X)/D_{\mathcal{V}} \longrightarrow \ker(\Delta_X - \lambda), \qquad [u] \longmapsto Y_{\mathcal{V}}(\lambda)u.$$

Moreover, by definition, $\ker \widetilde{\pi}_{\mathcal{V}^\#} = D_{\mathcal{V}}$. Since π_S is surjective, so is $\widetilde{\pi}_{\mathcal{V}^\#}$. Hence, by the first isomorphism theorem, it induces a linear isomorphism

$$\overline{\pi}_{\mathcal{V}^\#} : D_{\max}(\Delta_X)/D_{\mathcal{V}} \longrightarrow \mathcal{V}^\#, \qquad [u] \longmapsto \widetilde{\pi}_{\mathcal{V}^\#}(u).$$

We define the Poisson operator

$$\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) : \mathcal{V}^\# \longrightarrow \ker(\Delta_X - \lambda)$$

by $\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) := \overline{Y}_{\mathcal{V}}(\lambda) \circ \overline{\pi}^{-1}_{\mathcal{V}^\#}$. Equivalently, if $f \in \mathcal{V}^\#$ and $u \in D_{\max}(\Delta_X)$ satisfies $\widetilde{\pi}_{\mathcal{V}^\#}(u) = f$, then $\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f = Y_{\mathcal{V}}(\lambda)u$. Consequently, $Y_{\mathcal{V}}(\lambda) = \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) \circ \widetilde{\pi}_{\mathcal{V}^\#}$.

Definition 3.7. Let $\lambda \in \mathbb{C}$. A function $u_\lambda \in D_{\max}(\Delta_X)$ is called a formal solution of $\Delta_X u_\lambda = \lambda u_\lambda$ if $(\Delta_X - \lambda)u_\lambda = 0$ in the sense of distributions on X'. We denote the space of formal solutions at λ by $\ker(\Delta_X - \lambda)$. Throughout this section, this kernel is understood to be taken in $D_{\max}(\Delta_X)$.

Proposition 3.8. *Let $\lambda \in \rho(\Delta_{X,\mathcal{V}})$ and $f \in \mathcal{V}^\#$. A function $u_\lambda \in D_{\max}(\Delta_X)$ is a formal solution satisfying the boundary condition $\widetilde{\pi}_{\mathcal{V}^\#}(u_\lambda) = f$ if and only if $u_\lambda = \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$.*

Proof. Suppose that u_λ is a formal solution satisfying the boundary condition, and set $v_\lambda := \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$. By definition, $v_\lambda \in \ker(\Delta_X - \lambda)$ and $\widetilde{\pi}_{\mathcal{V}^\#}(v_\lambda) = f$. Hence $\widetilde{\pi}_{\mathcal{V}^\#}(u_\lambda - v_\lambda) = 0$, so $u_\lambda - v_\lambda \in D_{\mathcal{V}}$. Since both u_λ and v_λ are formal solutions, we have $(\Delta_{X,\mathcal{V}} - \lambda)(u_\lambda - v_\lambda) = 0$. Since $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, it follows that $u_\lambda - v_\lambda = 0$. Hence $u_\lambda = v_\lambda = \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$. The converse follows immediately from the definition of $\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)$. □

Thus $\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)$ parametrizes formal solutions uniquely by their $\mathcal{V}^\#$-boundary coordinates.

Definition 3.9. The Weyl function associated with the Lagrangian pair $(\mathcal{V}, \mathcal{V}^\#)$ is the operator-valued function $M^X_{\mathcal{V},\mathcal{V}^\#} : \rho(\Delta_{X,\mathcal{V}}) \to \operatorname{Hom}_{\mathbb{C}}(\mathcal{V}^\#, \mathcal{V})$ defined by

$$M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) := \widetilde{\pi}_{\mathcal{V}} \circ \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda).$$

Let $\lambda \in \rho(\Delta_{X,\mathcal{V}})$ and $f \in \mathcal{V}^\#$. Since $\widetilde{\pi}_{\mathcal{V}^\#}\big(\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f\big) = f$, the decomposition $V_S = \mathcal{V} \oplus \mathcal{V}^\#$ gives $\pi_S\big(\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f\big) = M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f + f$. Equivalently, if $u \in D_{\max}(\Delta_X)$ satisfies $\widetilde{\pi}_{\mathcal{V}^\#}(u) = f$, then $Y_{\mathcal{V}}(\lambda)u = \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$, and hence $\pi_S\big(Y_{\mathcal{V}}(\lambda)u\big) = M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f + f$.

Definition 3.10. Let $(\mathcal{V}, \mathcal{V}^\#)$ be a Lagrangian pair in (V_S, ω_S). For a linear map $A : \mathcal{V}^\# \to \mathcal{V}$, its symplectic adjoint is the unique linear map $A^{\dagger_{\omega_S}} : \mathcal{V}^\# \to \mathcal{V}$ satisfying $\omega_S(Af, g) = \omega_S(f, A^{\dagger_{\omega_S}} g)$ for all $f, g \in \mathcal{V}^\#$. Equivalently, $\omega_S(Af, g) = -\omega_S(A^{\dagger_{\omega_S}} g, f)$. Existence and uniqueness follow from the nondegeneracy of the pairing between $\mathcal{V}$ and $\mathcal{V}^\#$ induced by ω_S. When no confusion can arise, we write $A^\dagger$ for $A^{\dagger_{\omega_S}}$.

Lemma 3.11. *Let $(\mathcal{V}, \mathcal{V}^\#)$ be a Lagrangian pair in (V_S, ω_S). Then, for every $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, one has*

$$M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)^{\dagger_{\omega_S}} = -M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda}).$$

Consequently, if β is a basis of $\mathcal{V}$ and $\beta^\#$ is its ω_S-dual basis in $\mathcal{V}^\#$, then

$$[M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)]^\beta_{\beta^\#} = ([M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda})]^\beta_{\beta^\#})^*.$$

In particular, if $\lambda \in \rho(\Delta_{X,\mathcal{V}}) \cap \mathbb{R}$, then $[M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)]^\beta_{\beta^\#}$ is Hermitian.

Proof. Since $\Delta_{X,\mathcal{V}}$ is self-adjoint, $\overline{\lambda} \in \rho(\Delta_{X,\mathcal{V}})$. Let $f, g \in \mathcal{V}^\#$, and set $u_\lambda := \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) f$ and $v_{\overline{\lambda}} := \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda}) g$. Since $\Delta_X u_\lambda = \lambda u_\lambda$ and $\Delta_X v_{\overline{\lambda}} = \overline{\lambda} v_{\overline{\lambda}}$, one has $q_X(u_\lambda, v_{\overline{\lambda}}) = 0$.

The global Green formula, the definition of the Weyl function, and the fact that both $\mathcal{V}$ and $\mathcal{V}^\#$ are Lagrangian give

$$\omega_S\big(M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) f, g\big) = -\omega_S\big(f, M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda}) g\big).$$

Equivalently,

$$\omega_S\big(M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) f, g\big) = \omega_S\big(f, -M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda}) g\big).$$

By the definition of the symplectic adjoint, $M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)^{\dagger_{\omega_S}} = -M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda})$.

With respect to the ω_S-dual bases $\beta^\#$ and β, one has $[A^{\dagger_{\omega_S}}]^\beta_{\beta^\#} = -([A]^\beta_{\beta^\#})^*$. Applying this identity to $A = M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)$ gives $[M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda})]^\beta_{\beta^\#} = ([M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)]^\beta_{\beta^\#})^*$, which is equivalent to the asserted identity. If λ is real, it follows that $[M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)]^\beta_{\beta^\#}$ is Hermitian. □

For each $i \geq 0$, recall that $q_i^{\mathcal{V}} : D_{\max}(\Delta_X) \to \mathbb{C}$ is defined by $q_i^{\mathcal{V}}(u) := q_X(u, \phi_i^{\mathcal{V}})$. Since $\phi_i^{\mathcal{V}} \in D_{\mathcal{V}}$ and $\Delta_{X,\mathcal{V}}$ is self-adjoint, one has $q_i^{\mathcal{V}}(u) = 0$ for every $u \in D_{\mathcal{V}}$. Thus $q_i^{\mathcal{V}}$ induces a linear functional $\overline{q}_i^{\mathcal{V}} : D_{\max}(\Delta_X)/D_{\mathcal{V}} \to \mathbb{C}$, given by $\overline{q}_i^{\mathcal{V}}([u]) = q_i^{\mathcal{V}}(u)$.

Define $\omega_i^{\mathcal{V},\mathcal{V}^\#} : \mathcal{V}^\# \to \mathbb{C}$ by $\omega_i^{\mathcal{V},\mathcal{V}^\#} := \overline{q}_i^{\mathcal{V}} \circ \overline{\pi}^{-1}_{\mathcal{V}^\#}$. Equivalently, if $u \in D_{\max}(\Delta_X)$ satisfies $\widetilde{\pi}_{\mathcal{V}^\#}(u) = f$, then $\omega_i^{\mathcal{V},\mathcal{V}^\#}(f) = q_i^{\mathcal{V}}(u)$.

It follows from the eigenfunction expansion of $Y_{\mathcal{V}}(\lambda)$ and Parseval's identity that, for $\lambda, \mu \in \rho(\Delta_{X,\mathcal{V}})$ and $f, g \in \mathcal{V}^\#$,

$$\big\langle \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) f, \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\mu) g \big\rangle_{L^2(X')} = \sum_{i \geq 0} \frac{\omega_i^{\mathcal{V},\mathcal{V}^\#}(f) \overline{\omega_i^{\mathcal{V},\mathcal{V}^\#}(g)}}{(\lambda - \lambda_i^{\mathcal{V}})(\overline{\mu} - \lambda_i^{\mathcal{V}})}.$$

Since $\Delta_{X,\mathcal{V}}$ is self-adjoint, $\lambda \in \rho(\Delta_{X,\mathcal{V}})$ implies $\overline{\lambda} \in \rho(\Delta_{X,\mathcal{V}})$. Taking $\mu = \overline{\lambda}$, we obtain

$$\big\langle \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) f, \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda}) g \big\rangle_{L^2(X')} = \sum_{i \geq 0} \frac{\omega_i^{\mathcal{V},\mathcal{V}^\#}(f) \overline{\omega_i^{\mathcal{V},\mathcal{V}^\#}(g)}}{(\lambda - \lambda_i^{\mathcal{V}})^2}.$$

Lemma 3.12. *The Weyl function $M^X_{\mathcal{V},\mathcal{V}^\#}$ is holomorphic on $\rho(\Delta_{X,\mathcal{V}})$. For $f, g \in \mathcal{V}^\#$ and $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, one has*

$$\omega_S\left(\frac{d}{d\lambda} M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) f, g\right) = \big\langle \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) f, \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda}) g \big\rangle_{L^2(X')}.$$

Equivalently,

$$\omega_S\left(\frac{d}{d\lambda}M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f,g\right)=\sum_{i\geq 0}\frac{\omega_i^{\mathcal{V},\mathcal{V}^\#}(f)\overline{\omega_i^{\mathcal{V},\mathcal{V}^\#}(g)}}{(\lambda-\lambda_i^{\mathcal{V}})^2},$$

where $\omega_i^{\mathcal{V},\mathcal{V}^\#}:=\overline{q}_i^{\mathcal{V}}\circ\overline{\pi}_{\mathcal{V}^\#}^{-1}$.

Proof. Fix $f\in\mathcal{V}^\#$, and choose $w_f\in D_{\max}(\Delta_X)$ such that $\widetilde{\pi}_{\mathcal{V}^\#}(w_f)=f$. Then

$$\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f=Y_{\mathcal{V}}(\lambda)w_f=w_f+R_{\mathcal{V}}(\lambda)(\Delta_X-\lambda)w_f.$$

The resolvent $R_{\mathcal{V}}(\lambda)$ is holomorphic on $\rho(\Delta_{X,\mathcal{V}})$, with values in $D_{\mathcal{V}}$ equipped with the graph norm. Hence $\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$ is holomorphic as a $D_{\max}(\Delta_X)$-valued function. Since $\widetilde{\pi}_{\mathcal{V}}$ is continuous with respect to the graph norm, it follows that $M^X_{\mathcal{V},\mathcal{V}^\#}$ is holomorphic. Set $u_f(\lambda):=\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$ and $u_g(\overline{\lambda}):=\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda})g$. Then $(\Delta_X-\lambda)u_f(\lambda)=0$ and $(\Delta_X-\overline{\lambda})u_g(\overline{\lambda})=0$. Moreover, $\pi_S(u_f(\lambda))=M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f+f$. Differentiating with respect to λ, we obtain $\pi_S\big(u_f'(\lambda)\big)=\frac{d}{d\lambda}M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$ and $(\Delta_X-\lambda)u_f'(\lambda)=u_f(\lambda)$. Using the definition of q_X and the convention that the $L^2(X')$-inner product is linear in the first variable, we obtain

$$\begin{aligned}q_X\big(u_f'(\lambda),u_g(\overline{\lambda})\big)&=\big\langle\lambda u_f'(\lambda)+u_f(\lambda),u_g(\overline{\lambda})\big\rangle_{L^2(X')}\\&\quad-\big\langle u_f'(\lambda),\overline{\lambda}u_g(\overline{\lambda})\big\rangle_{L^2(X')}\\&=\big\langle u_f(\lambda),u_g(\overline{\lambda})\big\rangle_{L^2(X')}.\end{aligned}$$

On the other hand, the global Green formula gives

$$\begin{aligned}q_X\big(u_f'(\lambda),u_g(\overline{\lambda})\big)&=\omega_S\left(\frac{d}{d\lambda}M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f,M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda})g+g\right)\\&=\omega_S\left(\frac{d}{d\lambda}M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f,g\right),\end{aligned}$$

because both $\frac{d}{d\lambda}M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f$ and $M^X_{\mathcal{V},\mathcal{V}^\#}(\overline{\lambda})g$ belong to the Lagrangian subspace $\mathcal{V}$. Comparing the two expressions proves the first identity. The series representation follows from the Parseval identity established above. □

Corollary 3.13. *Let $(\mathcal{V},\mathcal{V}^\#)$ be a Lagrangian pair in (V_S,ω_S). Then the associated Weyl function $M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)$ extends meromorphically across the eigenvalues of $\Delta_{X,\mathcal{V}}$, and all its possible poles are simple.*

More precisely, let λ_0 be an eigenvalue of $\Delta_{X,\mathcal{V}}$. Then, near λ_0,

$$M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)=-\frac{B_{\lambda_0}}{\lambda-\lambda_0}+H_{\lambda_0}(\lambda),$$

where H_{λ_0} is holomorphic near λ_0, and $B_{\lambda_0}:\mathcal{V}^\#\to\mathcal{V}$ is uniquely determined by

$$\omega_S(B_{\lambda_0}f,g)=\sum_{\lambda_i^{\mathcal{V}}=\lambda_0}\omega_i^{\mathcal{V},\mathcal{V}^\#}(f)\overline{\omega_i^{\mathcal{V},\mathcal{V}^\#}(g)}$$

for all $f,g\in\mathcal{V}^\#$. In particular, λ_0 is an actual pole of $M^X_{\mathcal{V},\mathcal{V}^\#}$ if and only if $B_{\lambda_0}\neq 0$.

Proof. Let λ_0 be an eigenvalue of $\Delta_{X,\mathcal{V}}$. Since $\Delta_{X,\mathcal{V}}$ has compact resolvent, λ_0 is isolated and has finite multiplicity. By Lemma 3.12, the singular part at λ_0 of

$$\omega_S\left(\frac{d}{d\lambda}M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)f,g\right)$$

is

$$\frac{1}{(\lambda-\lambda_0)^2}\sum_{\lambda_i^{\mathcal{V}}=\lambda_0}\omega_i^{\mathcal{V},\mathcal{V}^\#}(f)\overline{\omega_i^{\mathcal{V},\mathcal{V}^\#}(g)}.$$

The remaining part of the spectral series converges locally uniformly near λ_0. Indeed, fix $\zeta\in\rho(\Delta_{X,\mathcal{V}})$ near λ_0. The eigenfunction expansion of the Poisson operator shows that

$$\left\{\frac{\omega_i^{\mathcal{V},\mathcal{V}^\#}(f)}{\zeta-\lambda_i^{\mathcal{V}}}\right\}_{i\geq 0},\qquad \left\{\frac{\omega_i^{\mathcal{V},\mathcal{V}^\#}(g)}{\zeta-\lambda_i^{\mathcal{V}}}\right\}_{i\geq 0}$$

belong to ℓ^2. The Cauchy–Schwarz inequality therefore gives the required locally uniform convergence after the terms satisfying $\lambda_i^{\mathcal{V}}=\lambda_0$ are removed. Hence the remaining part is holomorphic near λ_0.

The pairing $\omega_S|_{\mathcal{V}\times\mathcal{V}^\#}$ is nondegenerate. Therefore, there exists a unique linear map $B_{\lambda_0}:\mathcal{V}^\#\to\mathcal{V}$ satisfying the identity in the statement. It follows that

$$\frac{d}{d\lambda}M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)=\frac{B_{\lambda_0}}{(\lambda-\lambda_0)^2}+K_{\lambda_0}(\lambda),$$

where K_{λ_0} is holomorphic near λ_0. Consequently, the derivative of $M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)+B_{\lambda_0}/(\lambda-\lambda_0)$ extends holomorphically across λ_0. It follows that

$$M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)=-\frac{B_{\lambda_0}}{\lambda-\lambda_0}+H_{\lambda_0}(\lambda)$$

for some H_{λ_0} holomorphic near λ_0. Therefore, every possible pole is simple, and λ_0 is an actual pole precisely when $B_{\lambda_0}\neq 0$. □

4. Kreĭn's Formula

We derive Kreĭn's formula for the resolvent difference of two self-adjoint realizations and the resulting trace identity. Fix a Lagrangian pair $(\mathcal{W},\mathcal{W}^\#)$, and denote by $P_{\mathcal{W}}:V_S\to\mathcal{W}$ and $P_{\mathcal{W}^\#}:V_S\to\mathcal{W}^\#$ the corresponding projections. We take $\Delta_{X,\mathcal{W}}$ as the reference realization and express the formulas through the Weyl function of $(\mathcal{W},\mathcal{W}^\#)$.

Let $(\mathcal{V},\mathcal{V}^\#)$ be another Lagrangian pair. Define $A_{\mathcal{V}}:=P_{\mathcal{W}}|_{\mathcal{V}}:\mathcal{V}\to\mathcal{W}$ and $C_{\mathcal{V}}:=P_{\mathcal{W}^\#}|_{\mathcal{V}}:\mathcal{V}\to\mathcal{W}^\#$. Thus every $v\in\mathcal{V}$ decomposes as $v=A_{\mathcal{V}}v+C_{\mathcal{V}}v$.

Define the symplectic adjoints $C_{\mathcal{V}}^{\dagger_{\omega_S}}:\mathcal{W}\to\mathcal{V}^\#$ and $A_{\mathcal{V}}^{\dagger_{\omega_S}}:\mathcal{W}^\#\to\mathcal{V}^\#$ by $\omega_S(v,C_{\mathcal{V}}^{\dagger_{\omega_S}}g)=\omega_S(C_{\mathcal{V}}v,g)$ and $\omega_S(v,A_{\mathcal{V}}^{\dagger_{\omega_S}}h)=\omega_S(A_{\mathcal{V}}v,h)$ for $v\in\mathcal{V}$, $g\in\mathcal{W}$, and $h\in\mathcal{W}^\#$. These adjoints exist uniquely by the nondegeneracy of the pairings between complementary Lagrangian subspaces.

Let $g\in\mathcal{W}$ and $h\in\mathcal{W}^\#$. Since $\mathcal{V}$ is Lagrangian, $g+h\in\mathcal{V}$ if and only if $\omega_S(g+h,v)=0$ for every $v\in\mathcal{V}$. Writing $v=A_{\mathcal{V}}v+C_{\mathcal{V}}v$, we obtain $\omega_S(g+h,v)=$

$\omega_S(C_{\mathcal{V}}^{\dagger\omega_S} g + A_{\mathcal{V}}^{\dagger\omega_S} h, v)$. By the nondegeneracy of $\omega_S|_{\mathcal{V}^\# \times \mathcal{V}}$, it follows that $g + h \in \mathcal{V}$ if and only if $C_{\mathcal{V}}^{\dagger\omega_S} g + A_{\mathcal{V}}^{\dagger\omega_S} h = 0$. Consequently, $\mathcal{V} = \ker(C_{\mathcal{V}}^{\dagger\omega_S} P_{\mathcal{W}} + A_{\mathcal{V}}^{\dagger\omega_S} P_{\mathcal{W}^\#})$.

For $\lambda \in \rho(\Delta_{X,\mathcal{W}})$, define the linear map

$$T_{(\mathcal{V},\mathcal{V}^\#),(\mathcal{W},\mathcal{W}^\#)} : \mathcal{W}^\# \longrightarrow \mathcal{V}^\#$$

by

$$T_{(\mathcal{V},\mathcal{V}^\#),(\mathcal{W},\mathcal{W}^\#)}(\lambda) = C_{\mathcal{V}}^{\dagger\omega_S} M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) + A_{\mathcal{V}}^{\dagger\omega_S}.$$

Once both Lagrangian pairs have been fixed, we abbreviate this operator as $T_{\mathcal{V},\mathcal{W}}(\lambda)$.

Lemma 4.1. *If $\lambda \in \rho(\Delta_{X,\mathcal{V}}) \cap \rho(\Delta_{X,\mathcal{W}})$, then $T_{\mathcal{V},\mathcal{W}}(\lambda)$ is a linear isomorphism from $\mathcal{W}^\#$ onto $\mathcal{V}^\#$.*

Proof. Since $\dim_{\mathbb{C}} \mathcal{V}^\# = \dim_{\mathbb{C}} \mathcal{W}^\#$, it suffices to prove injectivity. Let $f \in \ker T_{\mathcal{V},\mathcal{W}}(\lambda)$. Then $C_{\mathcal{V}}^{\dagger\omega_S} M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f + A_{\mathcal{V}}^{\dagger\omega_S} f = 0$. By the boundary equation for $\mathcal{V}$, this implies $M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f + f \in \mathcal{V}$.

Let $u_f(\lambda) := \Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f$. Then $u_f(\lambda)$ is the formal solution at λ whose $\mathcal{W}^\#$-boundary coordinate is f. By the definition of the Weyl function, $\pi_S(u_f(\lambda)) = M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f + f$. Hence $\pi_S(u_f(\lambda)) \in \mathcal{V}$, and therefore $u_f(\lambda) \in D_{\mathcal{V}}$. Since $(\Delta_X - \lambda) u_f(\lambda) = 0$, we have $(\Delta_{X,\mathcal{V}} - \lambda) u_f(\lambda) = 0$. The assumption $\lambda \in \rho(\Delta_{X,\mathcal{V}})$ gives $u_f(\lambda) = 0$. Consequently, $f = \widetilde{\pi}_{\mathcal{W}^\#}(u_f(\lambda)) = 0$. Thus $T_{\mathcal{V},\mathcal{W}}(\lambda)$ is injective and hence an isomorphism. □

Theorem 4.2 (Kreĭn's resolvent formula)**.** *Let $(\mathcal{V}, \mathcal{V}^\#)$ and $(\mathcal{W}, \mathcal{W}^\#)$ be Lagrangian pairs. If $\lambda \in \rho(\Delta_{X,\mathcal{V}}) \cap \rho(\Delta_{X,\mathcal{W}})$, then*

$$R_{\mathcal{V}}(\lambda) - R_{\mathcal{W}}(\lambda) = -\Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) T_{\mathcal{V},\mathcal{W}}(\lambda)^{-1} C_{\mathcal{V}}^{\dagger\omega_S} \widetilde{\pi}_{\mathcal{W}} R_{\mathcal{W}}(\lambda).$$

Proof. Let $F \in L^2(X')$, and set $v_\lambda := R_{\mathcal{V}}(\lambda) F \in D_{\mathcal{V}}$ and $w_\lambda := R_{\mathcal{W}}(\lambda) F \in D_{\mathcal{W}}$. Set $u_\lambda := v_\lambda - w_\lambda$. Since $(\lambda - \Delta_{X,\mathcal{V}}) v_\lambda = F$ and $(\lambda - \Delta_{X,\mathcal{W}}) w_\lambda = F$, we have $(\Delta_X - \lambda) u_\lambda = 0$. Thus u_λ is a formal solution at λ.

Since $\lambda \in \rho(\Delta_{X,\mathcal{W}})$, there exists a unique $f \in \mathcal{W}^\#$ such that $u_\lambda = \Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f$. Consequently, $\pi_S(u_\lambda) = M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f + f$. Since $v_\lambda = w_\lambda + u_\lambda$, it follows that

$$\pi_S(v_\lambda) = \pi_S(w_\lambda) + M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f + f.$$

Now $\pi_S(w_\lambda) \in \mathcal{W}$, whereas $\pi_S(v_\lambda) \in \mathcal{V}$. Applying the boundary equation for $\mathcal{V}$ therefore gives $T_{\mathcal{V},\mathcal{W}}(\lambda) f = -C_{\mathcal{V}}^{\dagger\omega_S} \widetilde{\pi}_{\mathcal{W}}(w_\lambda)$. By Lemma 4.1, $T_{\mathcal{V},\mathcal{W}}(\lambda)$ is invertible, and hence

$$f = -T_{\mathcal{V},\mathcal{W}}(\lambda)^{-1} C_{\mathcal{V}}^{\dagger\omega_S} \widetilde{\pi}_{\mathcal{W}}(w_\lambda).$$

Substituting this identity into $u_\lambda = \Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) f$ and using $w_\lambda = R_{\mathcal{W}}(\lambda) F$ gives the asserted identity applied to F. Since $F \in L^2(X')$ is arbitrary, the operator identity follows. □

Thus the difference between the resolvents of two self-adjoint realizations is determined entirely by the finite-dimensional critical asymptotic boundary data and the corresponding Weyl function. In particular, $R_{\mathcal{V}}(\lambda) - R_{\mathcal{W}}(\lambda)$ is a finite-rank operator of rank at most $\dim_{\mathbb{C}} \mathcal{W}^\# = \frac{1}{2} \dim_{\mathbb{C}} V_S$, and is therefore trace class.

Lemma 4.3. *Let $\lambda \in \rho(\Delta_{X,\mathcal{W}})$. Then*

$$\widetilde{\pi}_{\mathcal{W}} R_{\mathcal{W}}(\lambda)\Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) = -\frac{d}{d\lambda} M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)$$

as linear maps from $\mathcal{W}^\#$ to $\mathcal{W}$.

Proof. Let $f \in \mathcal{W}^\#$, and set $u_f(\lambda) := \Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)f$. Then $(\Delta_X - \lambda)u_f(\lambda) = 0$ and $\pi_S(u_f(\lambda)) = M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)f + f$.

Differentiating with respect to λ, we obtain

$$(\Delta_X - \lambda)u_f'(\lambda) = u_f(\lambda)$$

and

$$\pi_S(u_f'(\lambda)) = \frac{d}{d\lambda} M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)f.$$

In particular, the boundary data of $u_f'(\lambda)$ belong to $\mathcal{W}$, so $u_f'(\lambda) \in D_{\mathcal{W}}$.

Since $(\lambda - \Delta_{X,\mathcal{W}})u_f'(\lambda) = -u_f(\lambda)$, it follows that $u_f'(\lambda) = -R_{\mathcal{W}}(\lambda)u_f(\lambda)$. Applying $\widetilde{\pi}_{\mathcal{W}}$ gives

$$\frac{d}{d\lambda} M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)f = -\widetilde{\pi}_{\mathcal{W}} R_{\mathcal{W}}(\lambda)\Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)f.$$

Since this holds for every $f \in \mathcal{W}^\#$, the result follows. □

Choose fixed bases of $\mathcal{W}^\#$ and $\mathcal{V}^\#$, and write $\det T_{\mathcal{V},\mathcal{W}}(\lambda)$ for the determinant of the corresponding matrix. Although this determinant depends on the chosen bases, its logarithmic derivative does not.

Corollary 4.4. *If $\lambda \in \rho(\Delta_{X,\mathcal{V}}) \cap \rho(\Delta_{X,\mathcal{W}})$, then*

$$\begin{aligned}\operatorname{Tr}_{L^2(X')}\big(R_{\mathcal{V}}(\lambda) - R_{\mathcal{W}}(\lambda)\big) &= \operatorname{Tr}_{\mathcal{W}^\#}\big(T_{\mathcal{V},\mathcal{W}}(\lambda)^{-1}T'_{\mathcal{V},\mathcal{W}}(\lambda)\big)\\ &= \frac{d}{d\lambda}\log\det T_{\mathcal{V},\mathcal{W}}(\lambda).\end{aligned}$$

Proof. The resolvent difference is finite rank, so its trace is well-defined. By Kreĭn's resolvent formula and the cyclicity of the trace for finite-rank factorizations,

$$\begin{aligned}\operatorname{Tr}_{L^2(X')}\big(R_{\mathcal{V}}(\lambda) - R_{\mathcal{W}}(\lambda)\big) &= -\operatorname{Tr}_{\mathcal{W}^\#}\big(T_{\mathcal{V},\mathcal{W}}(\lambda)^{-1}C_{\mathcal{V}}^{\dagger\omega_S}\widetilde{\pi}_{\mathcal{W}}R_{\mathcal{W}}(\lambda)\Gamma^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)\big)\\ &= \operatorname{Tr}_{\mathcal{W}^\#}\big(T_{\mathcal{V},\mathcal{W}}(\lambda)^{-1}C_{\mathcal{V}}^{\dagger\omega_S}\frac{d}{d\lambda}M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)\big),\end{aligned}$$

where Lemma 4.3 was used in the second equality. Since $T'_{\mathcal{V},\mathcal{W}}(\lambda) = C_{\mathcal{V}}^{\dagger\omega_S}\frac{d}{d\lambda}M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda)$, this proves the first identity.

The map $T_{\mathcal{V},\mathcal{W}}(\lambda)$ is holomorphic on $\rho(\Delta_{X,\mathcal{W}})$ and invertible on $\rho(\Delta_{X,\mathcal{V}}) \cap \rho(\Delta_{X,\mathcal{W}})$. Jacobi's formula therefore gives the second identity. □

Fix bases of $\mathcal{W}^\#$ and $\mathcal{V}^\#$. Define $d^X_{(\mathcal{V},\mathcal{V}^\#),(\mathcal{W},\mathcal{W}^\#)}(\lambda)$ to be the determinant of the matrix representing $T_{(\mathcal{V},\mathcal{V}^\#),(\mathcal{W},\mathcal{W}^\#)}(\lambda)$ with respect to these bases. Once the two Lagrangian pairs and the bases have been fixed, we abbreviate this function as $d^X_{\mathcal{V},\mathcal{W}}(\lambda) := \det T_{\mathcal{V},\mathcal{W}}(\lambda)$.

Changing either basis multiplies $d^X_{\mathcal{V},\mathcal{W}}$ by a nonzero constant independent of λ. Hence its logarithmic derivative and divisor are independent of the chosen bases.

Proposition 4.5. *The function $d^X_{\mathcal{V},\mathcal{W}}$ extends meromorphically to $\mathbb{C}$. If λ_0 is an eigenvalue of either $\Delta_{X,\mathcal{V}}$ or $\Delta_{X,\mathcal{W}}$, then*

$$\operatorname{ord}_{\lambda_0} d^X_{\mathcal{V},\mathcal{W}} = \dim_{\mathbb{C}} \ker(\Delta_{X,\mathcal{V}} - \lambda_0) - \dim_{\mathbb{C}} \ker(\Delta_{X,\mathcal{W}} - \lambda_0).$$

Proof. Since $T_{\mathcal{V},\mathcal{W}}$ is meromorphic, $d^X_{\mathcal{V},\mathcal{W}} = \det T_{\mathcal{V},\mathcal{W}}$ extends meromorphically to $\mathbb{C}$. Moreover, for $\lambda \in \rho(\Delta_{X,\mathcal{V}}) \cap \rho(\Delta_{X,\mathcal{W}})$, Corollary 4.4 gives

$$\operatorname{Tr}_{L^2(X')}\big(R_{\mathcal{V}}(\lambda) - R_{\mathcal{W}}(\lambda)\big) = \frac{d}{d\lambda} \log d^X_{\mathcal{V},\mathcal{W}}(\lambda).$$

Fix λ_0, and denote by $\Pi_{\mathcal{V},\lambda_0}$ and $\Pi_{\mathcal{W},\lambda_0}$ the orthogonal projections onto $\ker(\Delta_{X,\mathcal{V}} - \lambda_0)$ and $\ker(\Delta_{X,\mathcal{W}} - \lambda_0)$, respectively. If λ_0 is not an eigenvalue of one of the operators, take the corresponding projection to be zero. Since $R_{\mathcal{V}}(\lambda) = (\lambda - \Delta_{X,\mathcal{V}})^{-1}$, subtracting the two Laurent expansions gives

$$R_{\mathcal{V}}(\lambda) - R_{\mathcal{W}}(\lambda) = \frac{\Pi_{\mathcal{V},\lambda_0} - \Pi_{\mathcal{W},\lambda_0}}{\lambda - \lambda_0} + H(\lambda),$$

where H is a holomorphic finite-rank operator-valued function near λ_0.

Taking the trace and comparing the residue at λ_0 with that of $(d^X_{\mathcal{V},\mathcal{W}})'/d^X_{\mathcal{V},\mathcal{W}}$ gives

$$\operatorname{ord}_{\lambda_0} d^X_{\mathcal{V},\mathcal{W}} = \dim_{\mathbb{C}} \ker(\Delta_{X,\mathcal{V}} - \lambda_0) - \dim_{\mathbb{C}} \ker(\Delta_{X,\mathcal{W}} - \lambda_0).$$

□

5. A Comparison Formula for Zeta Determinants

We compare the positive-spectrum zeta determinants of self-adjoint realizations. Since these realizations need not be nonnegative, all nonpositive eigenvalues are omitted, producing a finite-dimensional correction factor. A relative logarithmic derivative formula, combined with Kreĭn's trace identity, gives the comparison in terms of $d^X_{\mathcal{V},\mathcal{W}}(\mu)$ and $E^X_{\mathcal{V},\mathcal{W}}(\mu)$, subject to the following standing analytic hypothesis.

Assumption 5.1 (Zeta-regularity hypothesis). For every pair of self-adjoint realizations to which the comparison formula is applied, the corresponding positive-spectrum zeta functions admit meromorphic continuations to a neighborhood of $s = 0$, are regular there, and depend differentiably on the spectral shift. We also assume that the relative contour representation has sufficient locally uniform trace-norm decay to justify differentiation with respect to the shift, integration by parts on truncated contours, and passage to the limit.

Remark. The individual zeta-regularity requirement in Assumption 5.1 holds for the Friedrichs realization in the standard conic heat-kernel framework; see, for example, [11]. It is not automatic for a general self-adjoint realization. General Lagrangian boundary conditions may produce nonstandard poles or logarithmic singularities of the spectral zeta function, possibly at $s = 0$; see [13]. Accordingly, the ordinary zeta-determinant comparison below is stated only for pairs of realizations satisfying Assumption 5.1. A treatment of arbitrary self-adjoint realizations would in general require a suitably modified determinant. Thus Assumption 5.1 is an independent analytic input, not a consequence of the finite-dimensional extension theory.

Let $A : D(A) \subset \mathcal{H} \to \mathcal{H}$ be a self-adjoint operator on a separable infinite-dimensional Hilbert space $\mathcal{H}$. Assume that A is bounded from below and has compact resolvent. Its spectrum is then discrete and consists of real eigenvalues of finite multiplicity. We enumerate them, with multiplicity, as $\lambda_1 \le \lambda_2 \le \cdots$, where $\lambda_k \to +\infty$. Assume in addition that $\sum_{\lambda_k>0} \lambda_k^{-s}$ converges for $\operatorname{Re} s$ sufficiently large. The positive-spectrum zeta function of A is then defined in this right half-plane by

$$\zeta_A(s) := \sum_{\lambda_k>0} \lambda_k^{-s}.$$

Thus all nonpositive eigenvalues are omitted. Let $P_A^+ : \mathcal{H} \to \mathcal{H}$ be the spectral projection onto the positive spectral subspace of A. For $\operatorname{Re} s$ sufficiently large, the Mellin transform gives

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \operatorname{Tr}_{\mathcal{H}}\big(e^{-tA} P_A^+\big)\, dt.$$

If A is nonnegative, then $P_A^+ = I - P_{\ker A}$, where $P_{\ker A} : \mathcal{H} \to \mathcal{H}$ is the orthogonal projection onto $\ker A$. In this case,

$$\zeta_A(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \operatorname{Tr}_{\mathcal{H}}\big(e^{-tA} - P_{\ker A}\big)\, dt.$$

Suppose that ζ_A admits a meromorphic continuation to a neighborhood of $s = 0$ and is regular there. The positive-spectrum zeta determinant of A is defined by

$$\operatorname{Det}_\zeta(A) := \exp\big(-\zeta_A'(0)\big).$$

Throughout this section, $\operatorname{Det}_\zeta(A)$ always refers to this positive-spectrum convention.

Now let $A : D(A) \subset \mathcal{H} \to \mathcal{H}$ and $B : D(B) \subset \mathcal{H} \to \mathcal{H}$ be self-adjoint operators satisfying the preceding assumptions. Define their relative zeta function by $\zeta_{(A,B)}(s) := \zeta_A(s) - \zeta_B(s)$, and define their relative zeta determinant by $\operatorname{Det}_\zeta(A, B) := \exp(-\zeta_{(A,B)}'(0))$. It follows immediately that $\operatorname{Det}_\zeta(A, B) = \operatorname{Det}_\zeta(A)/\operatorname{Det}_\zeta(B)$.

Write $R_A(z) := (z - A)^{-1}$ and $R_B(z) := (z - B)^{-1}$.

Let $\gamma_{A,B}^+$ be an admissible positively oriented spectral contour surrounding the positive spectra of A and B, while excluding their nonpositive spectra. Since the positive spectra are unbounded, this contour is understood as an unbounded spectral contour, or equivalently as the limit of suitable truncated contours. We choose the branch of z^{-s} on $\mathbb{C} \setminus (-\infty, 0]$.

Assume that $R_A(z) - R_B(z)$ is trace class along the contour and has sufficient decay at infinity for the following integral to converge when $\operatorname{Re} s$ is sufficiently large. Then

$$\zeta_{(A,B)}(s) = \frac{1}{2\pi i} \int_{\gamma_{A,B}^+} z^{-s} \operatorname{Tr}_{\mathcal{H}}\big(R_A(z) - R_B(z)\big)\, dz.$$

Suppose further that the meromorphic continuation of this contour integral is regular at $s = 0$. By definition,

$$\log \operatorname{Det}_\zeta(A, B) = -\left.\frac{d}{ds}\right|_{s=0} \left[\frac{1}{2\pi i} \int_{\gamma_{A,B}^+} z^{-s} \operatorname{Tr}_{\mathcal{H}}\big(R_A(z) - R_B(z)\big)\, dz\right]^{\mathrm{mer}}.$$

If differentiation under the continued integral is justified at $s = 0$, this becomes

$$\log \mathrm{Det}_\zeta(A, B) = \frac{1}{2\pi i} \int_{\gamma^+_{A,B}} \log z \ \mathrm{Tr}_{\mathcal{H}}\big(R_A(z) - R_B(z)\big)\, dz.$$

For a real parameter $\mu \in \rho(A) \cap \rho(B)$, consider the shifted pair $(A-\mu, B-\mu)$. Under the same assumptions,

$$\log \mathrm{Det}_\zeta(A-\mu, B-\mu) = \frac{1}{2\pi i} \int_{\gamma^+_\mu} \log z \ \mathrm{Tr}_{\mathcal{H}}\big((z-A+\mu)^{-1} - (z-B+\mu)^{-1}\big)\, dz,$$

where γ^+_μ surrounds the positive spectra of $A-\mu$ and $B-\mu$ and excludes their nonpositive spectra.

After shrinking the interval of μ, if necessary, we may assume that no eigenvalue of $A-\mu$ or $B-\mu$ crosses zero and choose the admissible contour γ^+ independently of μ. Set $Q_A(z,\mu) := (z-A+\mu)^{-1}$ and $Q_B(z,\mu) := (z-B+\mu)^{-1}$. Then $\partial_\mu Q_A(z,\mu) = \partial_z Q_A(z,\mu) = -Q_A(z,\mu)^2$, and similarly for B.

Differentiate first on the truncated contours and integrate by parts there. By the assumed trace-norm decay of $Q_A(z,\mu) - Q_B(z,\mu)$, the boundary terms at the truncation points tend to zero, locally uniformly for μ in the interval under consideration. Passing to the limit gives

$$\frac{d}{d\mu} \log \mathrm{Det}_\zeta(A-\mu, B-\mu) = -\frac{1}{2\pi i} \int_{\gamma^+} \frac{1}{z} \mathrm{Tr}_{\mathcal{H}}\big(Q_A(z,\mu) - Q_B(z,\mu)\big)\, dz.$$

Let $P^+_{A-\mu}$ and $P^+_{B-\mu}$ be the spectral projections onto the positive spectral subspaces of $A-\mu$ and $B-\mu$, respectively. Applying the holomorphic functional calculus to the relative resolvent on truncated contours and then passing to the limit under the standing trace-class and decay assumptions gives

$$\begin{aligned} &\frac{1}{2\pi i} \int_{\gamma^+} \frac{1}{z} \mathrm{Tr}_{\mathcal{H}}\big(Q_A(z,\mu) - Q_B(z,\mu)\big)\, dz \\ &\qquad = \mathrm{Tr}_{\mathcal{H}}\left((A-\mu)^{-1} P^+_{A-\mu} - (B-\mu)^{-1} P^+_{B-\mu}\right). \end{aligned}$$

Here only the relative difference on the right is asserted to be trace class; its two terms need not be trace class separately. Consequently,

$$\frac{d}{d\mu} \log \mathrm{Det}_\zeta(A-\mu, B-\mu) = -\,\mathrm{Tr}_{\mathcal{H}}\big((A-\mu)^{-1} P^+_{A-\mu} - (B-\mu)^{-1} P^+_{B-\mu}\big).$$

We denote the trace on the right-hand side by

$$\mathrm{Tr}^+\big((A-\mu)^{-1} - (B-\mu)^{-1}\big) := \mathrm{Tr}_{\mathcal{H}}\big((A-\mu)^{-1} P^+_{A-\mu} - (B-\mu)^{-1} P^+_{B-\mu}\big).$$

Thus

$$\frac{d}{d\mu} \log \mathrm{Det}_\zeta(A-\mu, B-\mu) = -\,\mathrm{Tr}^+\big((A-\mu)^{-1} - (B-\mu)^{-1}\big).$$

Let $P^-_{A-\mu}$ and $P^-_{B-\mu}$ denote the spectral projections onto the negative spectral subspaces of $A-\mu$ and $B-\mu$, respectively. Define $\mathrm{Tr}^-\big((A-\mu)^{-1} - (B-\mu)^{-1}\big)$ by

$$\mathrm{Tr}_{\mathcal{H}}\big((A-\mu)^{-1} P^-_{A-\mu} - (B-\mu)^{-1} P^-_{B-\mu}\big).$$

These operators have finite rank because A and B are bounded from below. Since $\mu \in \rho(A) \cap \rho(B)$, zero is not an eigenvalue of either shifted operator. Hence

$$\begin{aligned}\operatorname{Tr}^+\big((A-\mu)^{-1}-(B-\mu)^{-1}\big) &= \operatorname{Tr}_{\mathcal{H}}\big((A-\mu)^{-1}-(B-\mu)^{-1}\big)\\ &\quad - \operatorname{Tr}^-\big((A-\mu)^{-1}-(B-\mu)^{-1}\big).\end{aligned}$$

Let $I \subset \mathbb{R} \cap \rho(A) \cap \rho(B)$ be an open interval. For $\mu \in I$, define

$$E_{A,B}(\mu) := \frac{\displaystyle\prod_{\mu-\lambda_\ell^A>0} (\mu-\lambda_\ell^A)^{m(\lambda_\ell^A)}}{\displaystyle\prod_{\mu-\lambda_\ell^B>0} (\mu-\lambda_\ell^B)^{m(\lambda_\ell^B)}}.$$

The products are taken over the distinct eigenvalues satisfying the indicated inequalities. They are finite, and an empty product is understood to be 1. Since every factor is positive, $E_{A,B}(\mu) > 0$.

Since no eigenvalue crosses μ while μ remains in I, the sets of eigenvalues appearing in the products are constant on I. Therefore,

$$\frac{d}{d\mu}\log E_{A,B}(\mu) = \sum_{\mu-\lambda_\ell^A>0} \frac{m(\lambda_\ell^A)}{\mu-\lambda_\ell^A} - \sum_{\mu-\lambda_\ell^B>0} \frac{m(\lambda_\ell^B)}{\mu-\lambda_\ell^B}.$$

Theorem 5.2. *Let A and B be self-adjoint operators on $\mathcal{H}$, bounded from below and having compact resolvent. Let $I \subset \mathbb{R} \cap \rho(A) \cap \rho(B)$ be an open interval. Assume that, for every $\mu \in I$, the positive-spectrum zeta functions of $A-\mu$ and $B-\mu$ admit meromorphic continuations that are regular at $s=0$ and depend differentiably on μ near $s=0$. Assume also that $(A-\mu)^{-1}-(B-\mu)^{-1}$ is trace class for every $\mu \in I$. Finally, assume that the relative contour representation above holds with sufficient trace-norm decay, locally uniformly in μ, to justify differentiation with respect to μ, integration by parts on truncated contours, and passage to the limit.*

Then

$$\begin{aligned}\frac{d}{d\mu}\log \operatorname{Det}_\zeta(A-\mu,B-\mu) &= -\operatorname{Tr}_{\mathcal{H}}\big((A-\mu)^{-1}-(B-\mu)^{-1}\big) - \frac{d}{d\mu}\log E_{A,B}(\mu)\\ &= \operatorname{Tr}_{\mathcal{H}}\big(R_A(\mu)-R_B(\mu)\big) - \frac{d}{d\mu}\log E_{A,B}(\mu).\end{aligned}$$

We now apply the relative formula to self-adjoint realizations of the conic Laplacian. Let $(\mathcal{V},\mathcal{V}^\#)$ and $(\mathcal{W},\mathcal{W}^\#)$ be Lagrangian pairs in (V_S,ω_S). Let $I \subset \mathbb{R} \cap \rho(\Delta_{X,\mathcal{V}}) \cap \rho(\Delta_{X,\mathcal{W}})$ be an open interval, and set $E^X_{\mathcal{V},\mathcal{W}}(\mu) := E_{\Delta_{X,\mathcal{V}},\Delta_{X,\mathcal{W}}}(\mu)$ for $\mu \in I$.

We verify the operator-theoretic hypotheses of Theorem 5.2 in the present setting. The realizations $\Delta_{X,\mathcal{V}}$ and $\Delta_{X,\mathcal{W}}$ are bounded from below and have compact resolvent, while Kreĭn's resolvent formula shows that $R_{\mathcal{V}}(\mu)-R_{\mathcal{W}}(\mu)$ is finite rank. Since

$$(\Delta_{X,\mathcal{V}}-\mu)^{-1}-(\Delta_{X,\mathcal{W}}-\mu)^{-1} = -\big(R_{\mathcal{V}}(\mu)-R_{\mathcal{W}}(\mu)\big),$$

the difference of the shifted inverses is also finite rank and hence trace class. The remaining analytic hypotheses are precisely Assumption 5.1. Theorem 5.2 expresses the logarithmic derivative of the relative zeta determinant as the trace of the resolvent

difference minus the logarithmic derivative of $E^X_{\mathcal{V},\mathcal{W}}$. By Corollary 4.4, this resolvent trace is $\frac{d}{d\mu}\log d^X_{\mathcal{V},\mathcal{W}}(\mu)$. Therefore,

$$\frac{d}{d\mu}\log \mathrm{Det}_\zeta(\Delta_{X,\mathcal{V}}-\mu,\Delta_{X,\mathcal{W}}-\mu)=\frac{d}{d\mu}\log d^X_{\mathcal{V},\mathcal{W}}(\mu)-\frac{d}{d\mu}\log E^X_{\mathcal{V},\mathcal{W}}(\mu).$$

Equivalently,

$$\frac{d}{d\mu}\log\left(\frac{\mathrm{Det}_\zeta(\Delta_{X,\mathcal{V}}-\mu,\Delta_{X,\mathcal{W}}-\mu)E^X_{\mathcal{V},\mathcal{W}}(\mu)}{d^X_{\mathcal{V},\mathcal{W}}(\mu)}\right)=0.$$

Hence the quantity inside the logarithm is constant on I, and the desired comparison formula follows.

Theorem 5.3. *Assume Assumption 5.1. Let I be an open interval contained in $\mathbb{R}\setminus(\sigma(\Delta_{X,\mathcal{V}})\cup\sigma(\Delta_{X,\mathcal{W}}))$. Then there exists a nonzero constant $C_{\mathcal{V},\mathcal{W}}$, independent of $\mu\in I$, such that*

$$\frac{\mathrm{Det}_\zeta(\Delta_{X,\mathcal{V}}-\mu)}{\mathrm{Det}_\zeta(\Delta_{X,\mathcal{W}}-\mu)}=C_{\mathcal{V},\mathcal{W}}\frac{d^X_{\mathcal{V},\mathcal{W}}(\mu)}{E^X_{\mathcal{V},\mathcal{W}}(\mu)}.$$

The constant may depend on I, the Lagrangian pairs $(\mathcal{V},\mathcal{V}^\#)$ and $(\mathcal{W},\mathcal{W}^\#)$, and the bases used to define $d^X_{\mathcal{V},\mathcal{W}}$.

Remark. The extension-theoretic, compactness, and trace-class ingredients of the theorem have been established in the preceding sections. The conditional input is confined to the zeta continuation and relative contour estimates stated in Assumption 5.1.

Corollary 5.4. *Assume Assumption 5.1. Set $r:=\dim_{\mathbb{C}}\ker\Delta_{X,\mathcal{V}}-\dim_{\mathbb{C}}\ker\Delta_{X,\mathcal{W}}$, and define $(d^X_{\mathcal{V},\mathcal{W}})^{\mathrm{red}}(\lambda):=\lambda^{-r}d^X_{\mathcal{V},\mathcal{W}}(\lambda)$. Then $(d^X_{\mathcal{V},\mathcal{W}})^{\mathrm{red}}$ is holomorphic and nonzero at $\lambda=0$, and*

$$\frac{\mathrm{Det}_\zeta(\Delta_{X,\mathcal{V}})}{\mathrm{Det}_\zeta(\Delta_{X,\mathcal{W}})}=C_{\mathcal{V},\mathcal{W}}(d^X_{\mathcal{V},\mathcal{W}})^{\mathrm{red}}(0)\frac{\displaystyle\prod_{\lambda^{\mathcal{W}}_\ell<0}(-\lambda^{\mathcal{W}}_\ell)^{m(\lambda^{\mathcal{W}}_\ell)}}{\displaystyle\prod_{\lambda^{\mathcal{V}}_\ell<0}(-\lambda^{\mathcal{V}}_\ell)^{m(\lambda^{\mathcal{V}}_\ell)}}.$$

The products are taken over the distinct negative eigenvalues, with their multiplicities appearing as exponents, and an empty product is understood to be 1.

Proof. Choose $\varepsilon>0$ such that $(-\varepsilon,\varepsilon)\cap\big(\sigma(\Delta_{X,\mathcal{V}})\cup\sigma(\Delta_{X,\mathcal{W}})\big)\subset\{0\}$, and apply Theorem 5.3 on $I=(0,\varepsilon)$.

For $\mu\in I$, every zero eigenvalue becomes the negative eigenvalue $-\mu$ of the corresponding shifted operator. By the definition of $E^X_{\mathcal{V},\mathcal{W}}$, $E^X_{\mathcal{V},\mathcal{W}}(\mu)=\mu^r\widetilde{E}^X_{\mathcal{V},\mathcal{W}}(\mu)$, where

$$\widetilde{E}^X_{\mathcal{V},\mathcal{W}}(\mu):=\frac{\displaystyle\prod_{\lambda^{\mathcal{V}}_\ell<0}(\mu-\lambda^{\mathcal{V}}_\ell)^{m(\lambda^{\mathcal{V}}_\ell)}}{\displaystyle\prod_{\lambda^{\mathcal{W}}_\ell<0}(\mu-\lambda^{\mathcal{W}}_\ell)^{m(\lambda^{\mathcal{W}}_\ell)}}.$$

By Proposition 4.5, $\operatorname{ord}_0 d^X_{\mathcal{V},\mathcal{W}} = r$. Hence $d^X_{\mathcal{V},\mathcal{W}}(\mu) = \mu^r (d^X_{\mathcal{V},\mathcal{W}})^{\mathrm{red}}(\mu)$. The factors μ^r cancel in the shifted comparison formula, giving

$$\frac{\operatorname{Det}_\zeta(\Delta_{X,\mathcal{V}} - \mu)}{\operatorname{Det}_\zeta(\Delta_{X,\mathcal{W}} - \mu)} = C_{\mathcal{V},\mathcal{W}} \frac{(d^X_{\mathcal{V},\mathcal{W}})^{\mathrm{red}}(\mu)}{\widetilde{E}^X_{\mathcal{V},\mathcal{W}}(\mu)}.$$

Letting $\mu \to 0^+$, the positive-spectrum convention shows that the zeta determinants of the shifted operators converge to the corresponding unshifted positive-spectrum zeta determinants. Moreover,

$$\widetilde{E}^X_{\mathcal{V},\mathcal{W}}(0) = \frac{\displaystyle\prod_{\lambda^{\mathcal{V}}_\ell < 0} (-\lambda^{\mathcal{V}}_\ell)^{m(\lambda^{\mathcal{V}}_\ell)}}{\displaystyle\prod_{\lambda^{\mathcal{W}}_\ell < 0} (-\lambda^{\mathcal{W}}_\ell)^{m(\lambda^{\mathcal{W}}_\ell)}}.$$

The result follows. □

6. The Weyl Curve and the Determinant Line

We now organize the preceding analytic results geometrically. Related maps into Grassmannians, formed from null spaces of elliptic cone operators, appear in [9], while Wang developed an abstract theory of Weyl curves for symmetric operators [22]. The treatment below is self-contained and is formulated directly in terms of the critical asymptotic boundary data constructed above. Its purpose is to interpret the analytic identities from the preceding sections: local graph coordinates recover the Weyl functions, the real tangent form recovers their derivative identity, and the pullback determinant line gives a geometric interpretation of the boundary and zeta determinants.

For $\lambda \in \mathbb{C}$, define the intrinsic boundary-data subspace $\mathcal{K}_X(\lambda) := \pi_S\big(\ker(\Delta_X - \lambda)\big) \subset V_S$.

Lemma 6.1. *For every $\lambda \in \mathbb{C}$, one has $\mathcal{K}_X(\overline{\lambda}) \subset \mathcal{K}_X(\lambda)^{\perp_{\omega_S}}$. In particular, if $\lambda \in \mathbb{R}$, then $\mathcal{K}_X(\lambda)$ is isotropic.*

Proof. Let $g \in \mathcal{K}_X(\overline{\lambda})$ and $h \in \mathcal{K}_X(\lambda)$. Choose $v \in \ker(\Delta_X - \overline{\lambda})$ and $u \in \ker(\Delta_X - \lambda)$ such that $\pi_S v = g$ and $\pi_S u = h$. By the Green identity,

$$\begin{aligned} \omega_S(g,h) &= q_X(v,u) \\ &= \langle \Delta_X v, u\rangle - \langle v, \Delta_X u\rangle \\ &= \overline{\lambda}\langle v,u\rangle - \overline{\lambda}\langle v,u\rangle = 0. \end{aligned}$$

Here the second equality in the last line uses the convention that the inner product is linear in the first variable and conjugate linear in the second. Hence $g \in \mathcal{K}_X(\lambda)^{\perp_{\omega_S}}$, proving the asserted inclusion.

If $\lambda \in \mathbb{R}$, then $\overline{\lambda} = \lambda$, and hence $\mathcal{K}_X(\lambda) \subset \mathcal{K}_X(\lambda)^{\perp_{\omega_S}}$. Thus $\mathcal{K}_X(\lambda)$ is isotropic. □

Lemma 6.2. *Let $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$. If $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, then $\mathcal{K}_X(\overline{\lambda}) = \mathcal{K}_X(\lambda)^{\perp_{\omega_S}}$. In particular, if $\lambda \in \mathbb{R} \cap \rho(\Delta_{X,\mathcal{V}})$, then $\mathcal{K}_X(\lambda)$ is Lagrangian.*

Proof. By Lemma 6.1, $\mathcal{K}_X(\overline{\lambda}) \subset \mathcal{K}_X(\lambda)^{\perp_{\omega_S}}$. Choose a Lagrangian complement $\mathcal{V}^\#$ of $\mathcal{V}$. Since $\Delta_{X,\mathcal{V}}$ is self-adjoint, $\lambda \in \rho(\Delta_{X,\mathcal{V}})$ implies $\overline{\lambda} \in \rho(\Delta_{X,\mathcal{V}})$. For $\mu \in \{\lambda, \overline{\lambda}\}$, the Poisson operator $\Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\mu) : \mathcal{V}^\# \to \ker(\Delta_X - \mu)$ is an isomorphism. Moreover, the restriction of π_S to $\ker(\Delta_X - \mu)$ is injective. Indeed, if $u \in \ker(\Delta_X - \mu)$ and $\pi_S u = 0$, then $u \in D_{\min}(\Delta_X) \subset D_{\mathcal{V}}$, and hence $(\Delta_{X,\mathcal{V}} - \mu)u = 0$. Since $\mu \in \rho(\Delta_{X,\mathcal{V}})$, it follows that $u = 0$. By the definition of $\mathcal{K}_X(\mu)$, this restriction is also surjective onto $\mathcal{K}_X(\mu)$, and is therefore an isomorphism. Consequently,

$$\dim_{\mathbb{C}} \mathcal{K}_X(\lambda) = \dim_{\mathbb{C}} \mathcal{K}_X(\overline{\lambda}) = \dim_{\mathbb{C}} \mathcal{V}^\# = \frac{1}{2} \dim_{\mathbb{C}} V_S.$$

Since ω_S is nondegenerate, $\dim_{\mathbb{C}} \mathcal{K}_X(\lambda)^{\perp_{\omega_S}} = \dim_{\mathbb{C}} V_S - \dim_{\mathbb{C}} \mathcal{K}_X(\lambda) = \frac{1}{2} \dim_{\mathbb{C}} V_S$. The preceding inclusion is therefore an equality.

If $\lambda \in \mathbb{R}$, then $\overline{\lambda} = \lambda$, and hence $\mathcal{K}_X(\lambda) = \mathcal{K}_X(\lambda)^{\perp_{\omega_S}}$. Thus $\mathcal{K}_X(\lambda)$ is Lagrangian. □

Set $\Sigma_X := \bigcup_{\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)} \rho(\Delta_{X,\mathcal{V}})$.

Proposition 6.3. *We have*

$$\Sigma_X = \mathbb{C} \setminus \sigma_p(\Delta_{X,\min}),$$

where $\sigma_p(\Delta_{X,\min})$ denotes the point spectrum of the minimal Laplacian. Consequently, Σ_X is open and path-connected.

Proof. Since $\Sigma_X = \bigcup_{\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)} \rho(\Delta_{X,\mathcal{V}})$, and each resolvent set is open, the set Σ_X is open. Suppose first that $\lambda_0 \in \sigma_p(\Delta_{X,\min})$. Then there exists $0 \neq u_0 \in D_{\min}(\Delta_X)$ such that $(\Delta_{X,\min} - \lambda_0)u_0 = 0$. For every $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$, we have $D_{\min}(\Delta_X) \subset D_{\mathcal{V}}$, and hence $(\Delta_{X,\mathcal{V}} - \lambda_0)u_0 = 0$. Thus $\lambda_0 \in \sigma_p(\Delta_{X,\mathcal{V}})$ for every $\mathcal{V}$, so $\lambda_0 \notin \Sigma_X$. Therefore

$$\sigma_p(\Delta_{X,\min}) \subset \mathbb{C} \setminus \Sigma_X.$$

Conversely, suppose that $\lambda_0 \notin \sigma_p(\Delta_{X,\min})$. We show that $\lambda_0 \in \Sigma_X$. If $\lambda_0 \notin \mathbb{R}$, then $\lambda_0 \in \rho(\Delta_{X,\mathcal{V}})$ for every $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$, since each $\Delta_{X,\mathcal{V}}$ is self-adjoint. Hence $\lambda_0 \in \Sigma_X$. Suppose now that $\lambda_0 \in \mathbb{R}$. Then $\mathcal{K}_X(\lambda_0)$ is isotropic. Choose a Lagrangian subspace $\mathcal{W} \subset V_S$ containing $\mathcal{K}_X(\lambda_0)$, and choose a Lagrangian complement $\mathcal{W}^\#$ of $\mathcal{W}$. It follows that $\mathcal{K}_X(\lambda_0) \cap \mathcal{W}^\# = \{0\}$. We claim that λ_0 is not an eigenvalue of $\Delta_{X,\mathcal{W}^\#}$. Indeed, suppose that $u \in D_{\mathcal{W}^\#}$ satisfies $(\Delta_{X,\mathcal{W}^\#} - \lambda_0)u = 0$. Then $\pi_S u \in \mathcal{K}_X(\lambda_0) \cap \mathcal{W}^\# = \{0\}$. Hence $u \in D_{\min}(\Delta_X)$ and $(\Delta_{X,\min} - \lambda_0)u = 0$. Since $\lambda_0 \notin \sigma_p(\Delta_{X,\min})$, we conclude that $u = 0$. Thus $\lambda_0 \notin \sigma_p(\Delta_{X,\mathcal{W}^\#})$. Since $\Delta_{X,\mathcal{W}^\#}$ is self-adjoint with compact resolvent, its spectrum consists entirely of isolated eigenvalues of finite multiplicity. Therefore $\lambda_0 \in \rho(\Delta_{X,\mathcal{W}^\#})$, and hence $\lambda_0 \in \Sigma_X$. This proves

$$\Sigma_X = \mathbb{C} \setminus \sigma_p(\Delta_{X,\min}).$$

Finally, since $\Delta_{X,\min}$ is symmetric, $\sigma_p(\Delta_{X,\min}) \subset \mathbb{R}$. Moreover, for any fixed $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$,

$$\sigma_p(\Delta_{X,\min}) \subset \sigma(\Delta_{X,\mathcal{V}}).$$

Since $\Delta_{X,\mathcal{V}}$ has compact resolvent, its spectrum has no finite accumulation point. Consequently, the point spectrum of $\Delta_{X,\min}$ is a closed discrete subset of $\mathbb{R}$. Its complement in $\mathbb{C}$ is path-connected. Therefore Σ_X is path-connected. □

Set $m := \frac{1}{2}\dim_{\mathbb{C}} V_S$, and let $\mathrm{Gr}_m(V_S)$ be the complex Grassmannian. Its points are the m-dimensional complex linear subspaces of V_S, and its complex dimension is m^2.

We regard the Lagrangian Grassmannian $\mathrm{LGr}(V_S, \omega_S)$ as a subset of $\mathrm{Gr}_m(V_S)$.

For every $\lambda \in \Sigma_X$, there exists $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$ such that $\lambda \in \rho(\Delta_{X,\mathcal{V}})$. Choose a Lagrangian complement $\mathcal{V}^\#$ of $\mathcal{V}$. The Poisson operator is an isomorphism from $\mathcal{V}^\#$ onto $\ker(\Delta_X - \lambda)$. Hence $\dim_{\mathbb{C}} \ker(\Delta_X - \lambda) = m$. Moreover, the restriction

$$\pi_S\big|_{\ker(\Delta_X - \lambda)} : \ker(\Delta_X - \lambda) \longrightarrow \mathcal{K}_X(\lambda)$$

is a linear isomorphism. Indeed, it is surjective by the definition of $\mathcal{K}_X(\lambda)$. If $u \in \ker(\Delta_X - \lambda)$ and $\pi_S u = 0$, then $u \in D_{\min}(\Delta_X) \subset D_{\mathcal{V}}$, and hence $(\Delta_{X,\mathcal{V}} - \lambda)u = 0$. Since $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, it follows that $u = 0$. Therefore

$$\dim_{\mathbb{C}} \mathcal{K}_X(\lambda) = \dim_{\mathbb{C}} \ker(\Delta_X - \lambda) = m.$$

We thus obtain a map

$$\mathcal{K}_X : \Sigma_X \longrightarrow \mathrm{Gr}_m(V_S), \qquad \lambda \longmapsto \mathcal{K}_X(\lambda).$$

We next prove that $\mathcal{K}_X$ is holomorphic. We first establish the local graph representation needed for this purpose.

Let $E \in \mathrm{Gr}_m(V_S)$, and choose $F \in \mathrm{Gr}_m(V_S)$ such that $V_S = E \oplus F$. Consider the open subset

$$\mathcal{U}_E = \{\mathcal{L} \in \mathrm{Gr}_m(V_S) : \mathcal{L} \cap E = \{0\}\}.$$

Let $P_E : V_S \to E$ and $P_F : V_S \to F$ be the projections associated with this decomposition. For every $\mathcal{L} \in \mathcal{U}_E$, the restriction $P_F|_{\mathcal{L}} : \mathcal{L} \to F$ is an isomorphism. Define $A_{\mathcal{L}} := P_E \circ \big(P_F|_{\mathcal{L}}\big)^{-1} \in \mathrm{Hom}_{\mathbb{C}}(F, E)$. Then, for every $h \in F$, $\big(P_F|_{\mathcal{L}}\big)^{-1} h = A_{\mathcal{L}} h + h$, and hence

$$\mathcal{L} = \{A_{\mathcal{L}} h + h : h \in F\} = \mathrm{graph}_{(E,F)}(A_{\mathcal{L}}).$$

The correspondence

$$\kappa_{(E,F)} : \mathcal{U}_E \longrightarrow \mathrm{Hom}_{\mathbb{C}}(F, E), \qquad \mathcal{L} \longmapsto A_{\mathcal{L}},$$

defines a coordinate chart on $\mathrm{Gr}_m(V_S)$.

Lemma 6.4. *Let $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$. For every $\lambda \in \rho(\Delta_{X,\mathcal{V}})$,*

$$\mathcal{K}_X(\lambda) \cap \mathcal{V} = \{0\}.$$

Proof. Let $h \in \mathcal{K}_X(\lambda) \cap \mathcal{V}$. By the definition of $\mathcal{K}_X(\lambda)$, there exists $u_\lambda \in \ker(\Delta_X - \lambda)$ such that $\pi_S u_\lambda = h$. Since $h \in \mathcal{V}$, we have $u_\lambda \in D_{\mathcal{V}}$. Hence $(\Delta_{X,\mathcal{V}} - \lambda)u_\lambda = 0$. Since $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, the operator $\Delta_{X,\mathcal{V}} - \lambda$ is invertible. Therefore $u_\lambda = 0$, and consequently $h = \pi_S u_\lambda = 0$. □

Let $(\mathcal{V}, \mathcal{V}^\#)$ be a Lagrangian pair. If $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, then $\mathcal{K}_X(\lambda) \cap \mathcal{V} = \{0\}$, and hence $\mathcal{K}_X(\lambda) \in \mathcal{U}_{\mathcal{V}}$. By the definition of the Weyl function,

$$\mathcal{K}_X(\lambda) = \mathrm{graph}_{(\mathcal{V},\mathcal{V}^\#)}\big(M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)\big).$$

Consequently,

$$\kappa_{(\mathcal{V},\mathcal{V}^\#)}\big(\mathcal{K}_X(\lambda)\big) = M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) \in \mathrm{Hom}_{\mathbb{C}}(\mathcal{V}^\#, \mathcal{V}).$$

Proposition 6.5. *The map $\mathcal{K}_X : \Sigma_X \longrightarrow \mathrm{Gr}_m(V_S)$ is holomorphic.*

Proof. Let $\lambda_0 \in \Sigma_X$. By the definition of Σ_X, there exists $\mathcal{V} \in \mathrm{LGr}(V_S, \omega_S)$ such that $\lambda_0 \in \rho(\Delta_{X,\mathcal{V}})$. Choose a Lagrangian complement $\mathcal{V}^\#$ of $\mathcal{V}$.

For every $\lambda \in \rho(\Delta_{X,\mathcal{V}})$, the previous lemma gives $\mathcal{K}_X(\lambda) \cap \mathcal{V} = \{0\}$. Hence $\mathcal{K}_X\big(\rho(\Delta_{X,\mathcal{V}})\big) \subset \mathcal{U}_{\mathcal{V}}$. Consider the identity chart on $\rho(\Delta_{X,\mathcal{V}}) \subset \mathbb{C}$ and the Grassmannian chart $\kappa_{(\mathcal{V},\mathcal{V}^\#)} : \mathcal{U}_{\mathcal{V}} \longrightarrow \mathrm{Hom}_{\mathbb{C}}(\mathcal{V}^\#, \mathcal{V})$. The corresponding local coordinate representation of $\mathcal{K}_X$ is

$$\kappa_{(\mathcal{V},\mathcal{V}^\#)} \circ \mathcal{K}_X\big|_{\rho(\Delta_{X,\mathcal{V}})} = M^X_{\mathcal{V},\mathcal{V}^\#}.$$

Since $M^X_{\mathcal{V},\mathcal{V}^\#}$ is holomorphic on $\rho(\Delta_{X,\mathcal{V}})$, the map $\mathcal{K}_X$ is holomorphic at λ_0. Since λ_0 was arbitrary, $\mathcal{K}_X$ is holomorphic on Σ_X. □

Definition 6.6. The holomorphic map

$$\mathcal{K}_X : \Sigma_X \longrightarrow \mathrm{Gr}_m(V_S)$$

is called the Weyl curve of X.

Differentiating the local coordinate identity established above immediately gives the following description of the differential of the Weyl curve.

Proposition 6.7. *Let $\lambda_0 \in \Sigma_X$. The holomorphic differential of the Weyl curve at λ_0 is the complex-linear map*

$$d\mathcal{K}_X|_{\lambda_0} : T^{1,0}_{\lambda_0}\Sigma_X \longrightarrow T^{1,0}_{\mathcal{K}_X(\lambda_0)} \mathrm{Gr}_m(V_S).$$

Let $(\mathcal{V}, \mathcal{V}^\#)$ be a Lagrangian pair such that $\lambda_0 \in \rho(\Delta_{X,\mathcal{V}})$. Under the natural identification

$$T^{1,0}_{\mathcal{K}_X(\lambda_0)} \mathrm{Gr}_m(V_S) \simeq \mathrm{Hom}_{\mathbb{C}}\big(\mathcal{K}_X(\lambda_0), V_S/\mathcal{K}_X(\lambda_0)\big),$$

the tangent vector $d\mathcal{K}_X|_{\lambda_0}\big(\frac{\partial}{\partial\lambda}\big|_{\lambda_0}\big)$ is given by

$$\begin{aligned} &d\mathcal{K}_X|_{\lambda_0} \left(\frac{\partial}{\partial\lambda}\bigg|_{\lambda_0} \right) \big(M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda_0) h^\# + h^\# \big) \\ &= \left[\frac{d}{d\lambda} M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) h^\# \bigg|_{\lambda=\lambda_0} \right] \end{aligned}$$

for every $h^\# \in \mathcal{V}^\#$, where the brackets denote the class in $V_S/\mathcal{K}_X(\lambda_0)$.

Equivalently, in the graph-coordinate chart determined by $(\mathcal{V}, \mathcal{V}^\#)$,

$$d\kappa_{(\mathcal{V},\mathcal{V}^\#)}|_{\mathcal{K}_X(\lambda_0)} \left(d\mathcal{K}_X|_{\lambda_0} \left(\frac{\partial}{\partial\lambda}\bigg|_{\lambda_0} \right) \right) = \frac{d}{d\lambda} M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)\bigg|_{\lambda=\lambda_0}.$$

Proof. Fix $h^\# \in \mathcal{V}^\#$. Near λ_0, the vector $s_{h^\#}(\lambda) = M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) h^\# + h^\#$ belongs to $\mathcal{K}_X(\lambda)$. Differentiating at λ_0 gives

$$s'_{h^\#}(\lambda_0) = \frac{d}{d\lambda} M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) h^\#\bigg|_{\lambda=\lambda_0}.$$

Under the standard identification of the holomorphic tangent space of the Grassmannian with $\mathrm{Hom}_{\mathbb{C}}\big(\mathcal{K}_X(\lambda_0), V_S/\mathcal{K}_X(\lambda_0)\big)$, the corresponding tangent vector is obtained

by taking the class of $s'_{h^\#}(\lambda_0)$ modulo $\mathcal{K}_X(\lambda_0)$. This proves the first formula. The coordinate formula follows by differentiating $\kappa_{(\mathcal{V},\mathcal{V}^\#)} \circ \mathcal{K}_X = M^X_{\mathcal{V},\mathcal{V}^\#}$ at λ_0. □

We now study the transition between the local coordinate representations of the Weyl curve determined by two Lagrangian pairs $(\mathcal{V},\mathcal{V}^\#)$ and $(\mathcal{W},\mathcal{W}^\#)$ in (V_S,ω_S).

Let V, W, E, F be subspaces of V_S such that $V_S = V \oplus W = E \oplus F$. Every linear map $T : V_S \to V_S$ admits a block representation relative to the source decomposition $V_S = V \oplus W$ and the target decomposition $V_S = E \oplus F$. Namely, there exist linear maps $T_{11} : V \to E, T_{12} : W \to E, T_{21} : V \to F$, and $T_{22} : W \to F$ such that, for $v \in V$ and $w \in W$,

$$T(v+w) = \big(T_{11}v + T_{12}w\big) + \big(T_{21}v + T_{22}w\big),$$

where the first term belongs to E and the second belongs to F. We denote this block representation by

$$[T]^{(E,F)}_{(V,W)} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}.$$

Assume that $\dim_{\mathbb{C}} V = \dim_{\mathbb{C}} W = \dim_{\mathbb{C}} E = \dim_{\mathbb{C}} F = m$. Consider the transition map $\kappa_{(E,F)} \circ \kappa^{-1}_{(V,W)}$ between the graph-coordinate charts determined by the decompositions $V_S = V \oplus W$ and $V_S = E \oplus F$. Its domain is the open subset of $\mathrm{Hom}_{\mathbb{C}}(W,V)$ corresponding to $\mathcal{U}_V \cap \mathcal{U}_E$, and its values lie in $\mathrm{Hom}_{\mathbb{C}}(F,E)$.

Write

$$[\mathrm{id}_{V_S}]^{(E,F)}_{(V,W)} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

The standard transition formula for graph coordinates is

$$\big(\kappa_{(E,F)} \circ \kappa^{-1}_{(V,W)}\big)(L) = (AL+B)(CL+D)^{-1}.$$

Moreover,

$$\kappa_{(V,W)}\big(\mathcal{U}_V \cap \mathcal{U}_E\big) = \{L \in \mathrm{Hom}_{\mathbb{C}}(W,V) : CL + D : W \longrightarrow F \text{ is invertible}\}.$$

We therefore obtain the following transformation law for the local coordinate representations of the Weyl curve.

Theorem 6.8. *Let $(\mathcal{V},\mathcal{V}^\#)$ and $(\mathcal{W},\mathcal{W}^\#)$ be Lagrangian pairs. Let*

$$[\mathrm{id}_{V_S}]^{(\mathcal{W},\mathcal{W}^\#)}_{(\mathcal{V},\mathcal{V}^\#)} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

be the block representation of the identity map relative to the two Lagrangian decompositions. If $\lambda \in \rho(\Delta_{X,\mathcal{V}}) \cap \rho(\Delta_{X,\mathcal{W}})$, then $CM^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) + D : \mathcal{V}^\# \longrightarrow \mathcal{W}^\#$ is invertible, and

$$M^X_{\mathcal{W},\mathcal{W}^\#}(\lambda) = \big(AM^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) + B\big)\big(CM^X_{\mathcal{V},\mathcal{V}^\#}(\lambda) + D\big)^{-1}.$$

Proof. We argue directly from the definition of the Weyl function.

Let $h^\# \in \mathcal{V}^\#$, and set $u_\lambda = \Gamma^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)h^\#$. Then $\widetilde{\pi}_{\mathcal{V}^\#}(u_\lambda) = h^\#$, and

$$\pi_S(u_\lambda) = M^X_{\mathcal{V},\mathcal{V}^\#}(\lambda)h^\# + h^\#.$$

With respect to the decomposition $V_S = \mathcal{W} \oplus \mathcal{W}^{\#}$, its boundary components are

$$\widetilde{\pi}_{\mathcal{W}}(u_\lambda) = \big(AM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda) + B\big)h^{\#},$$
$$\widetilde{\pi}_{\mathcal{W}^{\#}}(u_\lambda) = \big(CM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda) + D\big)h^{\#}.$$

Set $k^{\#} = \big(CM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda) + D\big)h^{\#}$. Since $\lambda \in \rho(\Delta_{X,\mathcal{W}})$, the defining property of the Poisson operator implies that $u_\lambda = \Gamma^X_{\mathcal{W},\mathcal{W}^{\#}}(\lambda)k^{\#}$.

If $k^{\#} = 0$, then $u_\lambda = 0$, and hence $h^{\#} = \widetilde{\pi}_{\mathcal{V}^{\#}}(u_\lambda) = 0$. Therefore $CM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda) + D$ is injective. Since $\dim_{\mathbb{C}} \mathcal{V}^{\#} = \dim_{\mathbb{C}} \mathcal{W}^{\#} = m$, it is invertible.

By the definition of the Weyl function,

$$M^X_{\mathcal{W},\mathcal{W}^{\#}}(\lambda)k^{\#} = \big(AM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda) + B\big)h^{\#}.$$

Since this holds for every $h^{\#} \in \mathcal{V}^{\#}$, we obtain

$$M^X_{\mathcal{W},\mathcal{W}^{\#}}(\lambda)\big(CM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda) + D\big) = AM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda) + B.$$

The asserted transformation formula now follows from the invertibility established above. □

Set $\Sigma^{\mathbb{R}}_X := \Sigma_X \cap \mathbb{R}$. For every $\lambda \in \Sigma^{\mathbb{R}}_X$, Lemma 6.2 implies that $\mathcal{K}_X(\lambda)$ is Lagrangian. Hence the restriction

$$\mathcal{K}^{\mathbb{R}}_X := \mathcal{K}_X|_{\Sigma^{\mathbb{R}}_X} : \Sigma^{\mathbb{R}}_X \longrightarrow \mathrm{LGr}(V_S, \omega_S)$$

is smooth. We call $\mathcal{K}^{\mathbb{R}}_X$ the real Weyl curve. We now introduce local graph coordinates on $\mathrm{LGr}(V_S, \omega_S)$ and prove that the restriction of the real Weyl curve to each connected component of $\Sigma^{\mathbb{R}}_X$ is regular.

We equip $\mathrm{LGr}(V_S, \omega_S)$ with the subspace topology induced from $\mathrm{Gr}_m(V_S)$. Let $(\mathcal{V}, \mathcal{V}^{\#})$ be a Lagrangian pair, and define $\mathcal{U}^{\mathrm{Lag}}_{\mathcal{V}} := \mathcal{U}_{\mathcal{V}} \cap \mathrm{LGr}(V_S, \omega_S)$. Since $\mathcal{U}_{\mathcal{V}}$ is open in $\mathrm{Gr}_m(V_S)$, the set $\mathcal{U}^{\mathrm{Lag}}_{\mathcal{V}}$ is open in $\mathrm{LGr}(V_S, \omega_S)$ with respect to the induced topology. Equivalently,

$$\mathcal{U}^{\mathrm{Lag}}_{\mathcal{V}} = \{\mathcal{L} \in \mathrm{LGr}(V_S, \omega_S) : \mathcal{L} \cap \mathcal{V} = \{0\}\}.$$

For $M \in \mathrm{Hom}_{\mathbb{C}}(\mathcal{V}^{\#}, \mathcal{V})$, its symplectic adjoint $M^{\dagger} : \mathcal{V}^{\#} \to \mathcal{V}$ is defined by

$$\omega_S(Mh^{\#}, k^{\#}) = \omega_S(h^{\#}, M^{\dagger}k^{\#}), \qquad h^{\#}, k^{\#} \in \mathcal{V}^{\#}.$$

The map $M^{\dagger}$ is uniquely determined by the nondegeneracy of the pairing between $\mathcal{V}$ and $\mathcal{V}^{\#}$ induced by ω_S. Moreover,

$$\omega_S\big(Mh^{\#} + h^{\#}, Mk^{\#} + k^{\#}\big) = \omega_S\big(h^{\#}, (M^{\dagger} + M)k^{\#}\big).$$

Hence $\mathrm{graph}_{(\mathcal{V},\mathcal{V}^{\#})}(M)$ is Lagrangian if and only if $M^{\dagger} = -M$. Set

$$\mathrm{Herm}_{\omega_S}(\mathcal{V}^{\#}, \mathcal{V}) := \big\{M \in \mathrm{Hom}_{\mathbb{C}}(\mathcal{V}^{\#}, \mathcal{V}) : M^{\dagger} = -M\big\}.$$

This is a real vector subspace of $\mathrm{Hom}_{\mathbb{C}}(\mathcal{V}^{\#}, \mathcal{V})$. The Grassmannian graph chart determined by $V_S = \mathcal{V} \oplus \mathcal{V}^{\#}$ therefore restricts to the real coordinate chart

$$\kappa^{\mathrm{Lag}}_{(\mathcal{V},\mathcal{V}^{\#})} : \mathcal{U}^{\mathrm{Lag}}_{\mathcal{V}} \longrightarrow \mathrm{Herm}_{\omega_S}(\mathcal{V}^{\#}, \mathcal{V}),$$

given by $\kappa^{\mathrm{Lag}}_{(\mathcal{V},\mathcal{V}^{\#})}\big(\mathrm{graph}_{(\mathcal{V},\mathcal{V}^{\#})}(M)\big) = M$.

For every $\lambda \in \rho(\Delta_{X,\mathcal{V}}) \cap \Sigma_X^{\mathbb{R}}$, we have $\mathcal{K}_X^{\mathbb{R}}(\lambda) \in \mathcal{U}_{\mathcal{V}}^{\mathrm{Lag}}$ and $\kappa_{(\mathcal{V},\mathcal{V}^{\#})}^{\mathrm{Lag}}\big(\mathcal{K}_X^{\mathbb{R}}(\lambda)\big) = M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)$. Thus the Weyl function is the local coordinate representation of the real Weyl curve in the Lagrangian graph chart determined by $(\mathcal{V},\mathcal{V}^{\#})$. In particular,

$$\big(M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)\big)^{\dagger} = -M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda).$$

Let $\mathcal{L} : I \to \mathrm{LGr}(V_S,\omega_S)$ be a smooth curve. Its tangent vector at λ is naturally identified with a Hermitian form $\mathfrak{q}_\lambda$ on $\mathcal{L}(\lambda)$, called the tangent form of the curve. More explicitly, suppose that, relative to a fixed Lagrangian pair $(\mathcal{V},\mathcal{V}^{\#})$, the curve is represented by $\mathcal{L}(\lambda) = \big\{M(\lambda)h^{\#} + h^{\#} : h^{\#} \in \mathcal{V}^{\#}\big\}$. Then

$$\mathfrak{q}_\lambda\big(M(\lambda)h^{\#} + h^{\#}, M(\lambda)k^{\#} + k^{\#}\big) = \omega_S\big(\dot{M}(\lambda)h^{\#}, k^{\#}\big)$$

for $h^{\#}, k^{\#} \in \mathcal{V}^{\#}$. The curve is called positive if $\mathfrak{q}_\lambda(v,v) > 0$ for every $0 \neq v \in \mathcal{L}(\lambda)$ and every $\lambda \in I$.

Proposition 6.9 (Positivity of the real Weyl curve)**.** *Let $(\mathcal{V},\mathcal{V}^{\#})$ be a Lagrangian pair, and let $I \subset \rho(\Delta_{X,\mathcal{V}}) \cap \Sigma_X^{\mathbb{R}}$ be an open interval. Then the restriction*

$$\mathcal{K}_X^{\mathbb{R}}|_I : I \longrightarrow \mathrm{LGr}(V_S,\omega_S)$$

is a positive curve. Its tangent form satisfies

$$\mathfrak{q}_\lambda\big(M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#} + h^{\#}, M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)k^{\#} + k^{\#}\big) = \big\langle \Gamma_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#}, \Gamma_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)k^{\#}\big\rangle_{L^2(X')}$$

for $h^{\#}, k^{\#} \in \mathcal{V}^{\#}$. In particular,

$$\mathfrak{q}_\lambda\big(M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#} + h^{\#}, M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#} + h^{\#}\big) = \big\|\Gamma_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#}\big\|_{L^2(X')}^2 > 0$$

for every $h^{\#} \neq 0$.

Proof. By the definition of the tangent form,

$$\mathfrak{q}_\lambda\big(M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#} + h^{\#}, M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)k^{\#} + k^{\#}\big) = \omega_S\left(\frac{d}{d\lambda}M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#}, k^{\#}\right).$$

By Lemma 3.12,

$$\omega_S\left(\frac{d}{d\lambda}M_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#}, k^{\#}\right) = \big\langle \Gamma_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)h^{\#}, \Gamma_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)k^{\#}\big\rangle_{L^2(X')}.$$

The desired formula follows immediately. Taking $k^{\#} = h^{\#} \neq 0$, and using the fact that $\Gamma_{\mathcal{V},\mathcal{V}^{\#}}^X(\lambda)$ is an isomorphism, gives positivity. □

We next pull back the tautological bundle on $\mathrm{Gr}_m(V_S)$ along the Weyl curve. Scott and Wojciechowski studied determinant lines over Grassmannians of self-adjoint elliptic boundary conditions and related their canonical sections to zeta determinants and boundary Fredholm determinants [19, 20]; relative zeta metrics on determinant lines were further considered in [18]. A related use of the determinant of the tautological bundle in connection with Schubert divisors appears in Wang's abstract setting [22, Section 10.1]. The identification made here is specific to the present conic problem: the base is the spectral-parameter curve $\mathcal{K}_X$, the boundary determinants arising from the resolvent trace identity become transition functions of its pullback determinant line, and the corrected zeta determinants become twisted local representatives.

We recall the local bundle coordinates only to fix the direction of these transition functions and to identify them precisely with $d^X_{\mathcal{V},\mathcal{W}}$.

Let $\mathcal{E} \to \mathrm{Gr}_m(V_S)$ be the tautological vector bundle, whose fiber over $\mathcal{L} \in \mathrm{Gr}_m(V_S)$ is $\mathcal{L}$ itself. We first describe its local trivializations and transition functions.

Let $E \in \mathrm{Gr}_m(V_S)$, and choose $F \in \mathrm{Gr}_m(V_S)$ such that $V_S = E \oplus F$. Recall that the graph-coordinate chart $\kappa_{(E,F)} : \mathcal{U}_E \to \mathrm{Hom}_{\mathbb{C}}(F, E)$ is defined on $\mathcal{U}_E = \{\mathcal{L} \in \mathrm{Gr}_m(V_S) : \mathcal{L} \cap E = \{0\}\}$. For $\mathcal{L} \in \mathcal{U}_E$, write $A_{\mathcal{L}} = \kappa_{(E,F)}(\mathcal{L})$, so that $\mathcal{L} = \{A_{\mathcal{L}}h + h : h \in F\}$.

Define

$$\tau_{(E,F)} : \mathcal{U}_E \times F \longrightarrow \mathcal{E}|_{\mathcal{U}_E}, \qquad \tau_{(E,F)}(\mathcal{L}, h) = (\mathcal{L}, A_{\mathcal{L}}h + h).$$

For each $\mathcal{L} \in \mathcal{U}_E$, the map $h \mapsto A_{\mathcal{L}}h + h$ is a linear isomorphism from F onto $\mathcal{L}$. Hence $\tau_{(E,F)}$ is a holomorphic local trivialization of $\mathcal{E}$ over $\mathcal{U}_E$. Its inverse is given by $\tau^{-1}_{(E,F)}(\mathcal{L}, \xi) = (\mathcal{L}, P_F\xi)$, where $P_F : V_S \to F$ is the projection associated with $V_S = E \oplus F$. Now suppose that $V_S = V \oplus W = E \oplus F$, and write

$$[\mathrm{id}_{V_S}]^{(E,F)}_{(V,W)} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

Let $\mathcal{L} \in \mathcal{U}_V \cap \mathcal{U}_E$, and set $L = \kappa_{(V,W)}(\mathcal{L})$. Since $\mathcal{L} = \{Lh + h : h \in W\}$, the F-component of $Lh + h$ is $(CL + D)h$. Therefore

$$\big(\tau^{-1}_{(E,F)} \circ \tau_{(V,W)}\big)(\mathcal{L}, h) = \big(\mathcal{L}, (CL + D)h\big).$$

Thus the transition function of the tautological vector bundle $\mathcal{E}$ on $\mathcal{U}_V \cap \mathcal{U}_E$ is

$$g_{(E,F),(V,W)} : \mathcal{U}_V \cap \mathcal{U}_E \longrightarrow \mathrm{Iso}_{\mathbb{C}}(W, F),$$

given by $g_{(E,F),(V,W)}(\mathcal{L}) = C\,\kappa_{(V,W)}(\mathcal{L}) + D$. The determinant line bundle of $\mathcal{E}$ is $\mathrm{Det}(\mathcal{E}) := \bigwedge^m \mathcal{E}$, whose fiber over $\mathcal{L} \in \mathrm{Gr}_m(V_S)$ is $\mathrm{Det}(\mathcal{L}) := \bigwedge^m \mathcal{L}$. Its transition functions are obtained by taking the top exterior powers of the transition functions of $\mathcal{E}$. Thus the transition function of $\mathrm{Det}(\mathcal{E})$ on $\mathcal{U}_V \cap \mathcal{U}_E$ is

$$\bigwedge^m g_{(E,F),(V,W)} : \mathcal{U}_V \cap \mathcal{U}_E \longrightarrow \mathrm{Iso}_{\mathbb{C}}\left(\bigwedge^m W, \bigwedge^m F\right),$$

given by

$$\left(\bigwedge^m g_{(E,F),(V,W)}\right)(\mathcal{L}) = \bigwedge^m \big(C\,\kappa_{(V,W)}(\mathcal{L}) + D\big).$$

After choosing nonzero elements of $\bigwedge^m W$ and $\bigwedge^m F$, this transition function is represented by the scalar $\det\big(C\,\kappa_{(V,W)}(\mathcal{L}) + D\big)$.

Pulling back $\mathcal{E}$ along the Weyl curve gives the holomorphic vector bundle $\mathcal{E}_X := \mathcal{K}_X^*\mathcal{E} \to \Sigma_X$, whose fiber at $\lambda \in \Sigma_X$ is naturally identified with the boundary-data subspace $\mathcal{K}_X(\lambda) \subset V_S$. Since the determinant construction commutes naturally with pullback, there is a canonical isomorphism

$$\mathrm{Det}(\mathcal{E}_X) = \mathrm{Det}(\mathcal{K}_X^*\mathcal{E}) \simeq \mathcal{K}_X^* \,\mathrm{Det}(\mathcal{E}).$$

Consequently, the fiber of $\mathrm{Det}(\mathcal{E}_X)$ at λ is $\bigwedge^m \mathcal{K}_X(\lambda)$.

Whenever several Lagrangian pairs are considered simultaneously, we choose once and for all a nonzero frame of $\bigwedge^m \mathcal{V}^{\#}$ for each pair $(\mathcal{V}, \mathcal{V}^{\#})$, and use these fixed frames in every pairwise determinant. With this convention, the resulting scalar transition functions satisfy the cocycle identities exactly.

Let $(\mathcal{V},\mathcal{V}^{\#})$ be a Lagrangian pair. Since $\mathcal{K}_X\big(\rho(\Delta_{X,\mathcal{V}})\big)\subset\mathcal{U}_{\mathcal{V}}$, the local trivialization $\tau_{(\mathcal{V},\mathcal{V}^{\#})}$ of $\mathcal{E}$ pulls back to the local trivialization

$$\tau^X_{\mathcal{V},\mathcal{V}^{\#}}:\rho(\Delta_{X,\mathcal{V}})\times\mathcal{V}^{\#}\longrightarrow\mathcal{E}_X|_{\rho(\Delta_{X,\mathcal{V}})},$$

given by $\tau^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda,h^{\#})=\big(\lambda,M^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda)h^{\#}+h^{\#}\big)$. In particular, for every fixed $h^{\#}\in\mathcal{V}^{\#}$, the map $s_{h^{\#}}(\lambda)=M^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda)h^{\#}+h^{\#}$ defines a local holomorphic section of $\mathcal{E}_X$.

Let $(\mathcal{W},\mathcal{W}^{\#})$ be another Lagrangian pair, and write

$$[\mathrm{id}_{V_S}]^{(\mathcal{W},\mathcal{W}^{\#})}_{(\mathcal{V},\mathcal{V}^{\#})}=\begin{bmatrix}A & B\\ C & D\end{bmatrix}.$$

On $\rho(\Delta_{X,\mathcal{V}})\cap\rho(\Delta_{X,\mathcal{W}})$, the transition function of $\mathcal{E}_X$ from the $(\mathcal{V},\mathcal{V}^{\#})$-trivialization to the $(\mathcal{W},\mathcal{W}^{\#})$-trivialization is

$$g^X_{(\mathcal{W},\mathcal{W}^{\#}),(\mathcal{V},\mathcal{V}^{\#})}:\rho(\Delta_{X,\mathcal{V}})\cap\rho(\Delta_{X,\mathcal{W}})\longrightarrow\mathrm{Iso}_{\mathbb{C}}(\mathcal{V}^{\#},\mathcal{W}^{\#}),$$

given by $g^X_{(\mathcal{W},\mathcal{W}^{\#}),(\mathcal{V},\mathcal{V}^{\#})}(\lambda)=CM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda)+D$. Taking the top exterior power gives the transition function of $\mathrm{Det}(\mathcal{E}_X)$:

$$\bigwedge^m g^X_{(\mathcal{W},\mathcal{W}^{\#}),(\mathcal{V},\mathcal{V}^{\#})}(\lambda)=\bigwedge^m\big(CM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda)+D\big).$$

After choosing nonzero elements of $\bigwedge^m\mathcal{V}^{\#}$ and $\bigwedge^m\mathcal{W}^{\#}$ as local frames, the corresponding scalar transition function from the $(\mathcal{V},\mathcal{V}^{\#})$-trivialization to the $(\mathcal{W},\mathcal{W}^{\#})$-trivialization is

$$g^{\mathrm{Det},X}_{(\mathcal{W},\mathcal{W}^{\#}),(\mathcal{V},\mathcal{V}^{\#})}(\lambda)=\det\big(CM^X_{\mathcal{V},\mathcal{V}^{\#}}(\lambda)+D\big).$$

On the other hand, the transition map in the opposite direction is

$$\begin{aligned}g^X_{(\mathcal{V},\mathcal{V}^{\#}),(\mathcal{W},\mathcal{W}^{\#})}(\lambda)&=\Big(g^X_{(\mathcal{W},\mathcal{W}^{\#}),(\mathcal{V},\mathcal{V}^{\#})}(\lambda)\Big)^{-1}\\&=C^{\dagger}_{\mathcal{V}}M^X_{\mathcal{W},\mathcal{W}^{\#}}(\lambda)+A^{\dagger}_{\mathcal{V}}.\end{aligned}$$

Therefore its scalar determinant is

$$\begin{aligned}g^{\mathrm{Det},X}_{(\mathcal{V},\mathcal{V}^{\#}),(\mathcal{W},\mathcal{W}^{\#})}(\lambda)&=\det\Big(C^{\dagger}_{\mathcal{V}}M^X_{\mathcal{W},\mathcal{W}^{\#}}(\lambda)+A^{\dagger}_{\mathcal{V}}\Big)\\&=d^X_{(\mathcal{V},\mathcal{V}^{\#}),(\mathcal{W},\mathcal{W}^{\#})}(\lambda).\end{aligned}$$

Thus the boundary determinant $d^X_{\mathcal{V},\mathcal{W}}$ is precisely the scalar transition function of the pullback determinant line bundle $\mathrm{Det}(\mathcal{E}_X)$ from the $(\mathcal{W},\mathcal{W}^{\#})$-trivialization to the $(\mathcal{V},\mathcal{V}^{\#})$-trivialization.

We now restrict the preceding construction to the real Weyl curve. Set $\mathcal{E}^{\mathbb{R}}_X:=\mathcal{E}_X|_{\Sigma^{\mathbb{R}}_X}$. Equivalently, $\mathcal{E}^{\mathbb{R}}_X=(\mathcal{K}^{\mathbb{R}}_X)^*\big(\mathcal{E}|_{\mathrm{LGr}(V_S,\omega_S)}\big)$. Its fiber at $\mu\in\Sigma^{\mathbb{R}}_X$ is the Lagrangian boundary-data subspace $\mathcal{K}^{\mathbb{R}}_X(\mu)$. The associated determinant line bundle is $\mathrm{Det}(\mathcal{E}^{\mathbb{R}}_X)=\mathrm{Det}(\mathcal{E}_X)|_{\Sigma^{\mathbb{R}}_X}$. For two Lagrangian pairs $(\mathcal{V},\mathcal{V}^{\#})$ and $(\mathcal{W},\mathcal{W}^{\#})$, the scalar transition function from the $(\mathcal{W},\mathcal{W}^{\#})$-trivialization to the $(\mathcal{V},\mathcal{V}^{\#})$-trivialization on $\rho(\Delta_{X,\mathcal{V}})\cap\rho(\Delta_{X,\mathcal{W}})\cap\mathbb{R}$ is $d^X_{\mathcal{V},\mathcal{W}}(\mu)$. Thus the boundary determinants appearing in the comparison formula are naturally interpreted as the transition functions of the determinant line bundle induced by the real Weyl curve.

Let I be a connected component of $\Sigma_X^{\mathbb{R}}$. Fix a collection $\mathfrak{A}$ of Lagrangian pairs such that the nonempty sets below cover I, and suppose that Assumption 5.1 holds for every pair of realizations indexed by $\mathfrak{A}$ on each connected component of their overlap. For $(\mathcal{V}, \mathcal{V}^{\#}) \in \mathfrak{A}$, set $I_{\mathcal{V}} := I \cap \rho(\Delta_{X,\mathcal{V}})$. The nonempty sets $I_{\mathcal{V}}$ form an open cover $\mathfrak{U} = \{I_{\mathcal{V}} : (\mathcal{V}, \mathcal{V}^{\#}) \in \mathfrak{A},\ I_{\mathcal{V}} \neq \varnothing\}$ of I. For $\mu \in I_{\mathcal{V}}$, define the individual correction factor

$$E_{\mathcal{V}}^X(\mu) := \prod_{\lambda_\ell^{\mathcal{V}} < \mu} (\mu - \lambda_\ell^{\mathcal{V}})^{m(\lambda_\ell^{\mathcal{V}})}.$$

The product is finite, and an empty product is understood to be 1. On $I_{\mathcal{V}} \cap I_{\mathcal{W}}$, one then has $E_{\mathcal{V},\mathcal{W}}^X = E_{\mathcal{V}}^X / E_{\mathcal{W}}^X$.

For pairs in $\mathfrak{A}$, on every nonempty overlap $I_{\mathcal{V}} \cap I_{\mathcal{W}}$, Theorem 5.3, applied on each connected component of the overlap, gives

$$\frac{\mathrm{Det}_\zeta(\Delta_{X,\mathcal{V}} - \mu)}{\mathrm{Det}_\zeta(\Delta_{X,\mathcal{W}} - \mu)} = C_{\mathcal{V},\mathcal{W}}(\mu) \frac{E_{\mathcal{W}}^X(\mu)}{E_{\mathcal{V}}^X(\mu)} d_{\mathcal{V},\mathcal{W}}^X(\mu),$$

where $C_{\mathcal{V},\mathcal{W}}$ is a locally constant $\mathbb{C}^\times$-valued function on the overlap.

Because the fixed determinant frames specified above are used throughout, on every nonempty triple overlap the multiplicativity of relative zeta determinants and correction factors, together with the cocycle identity $d_{\mathcal{V},\mathcal{W}}^X d_{\mathcal{W},\mathcal{U}}^X = d_{\mathcal{V},\mathcal{U}}^X$, gives $C_{\mathcal{V},\mathcal{U}} = C_{\mathcal{V},\mathcal{W}} C_{\mathcal{W},\mathcal{U}}$. Thus

$$\{C_{\mathcal{V},\mathcal{W}}\} \in \check{Z}^1\big(\mathfrak{U}, \underline{\mathbb{C}^\times}\big).$$

This cocycle determines a rank-one complex local system, or equivalently a flat complex line bundle, $\mathcal{L}_C \to I$. Since I is an open interval, $\mathcal{L}_C$ is trivializable, although no canonical trivialization is specified.

Define $\widehat{D}_{\mathcal{V}} : I_{\mathcal{V}} \to \mathbb{C}^\times$ by $\widehat{D}_{\mathcal{V}}(\mu) := E_{\mathcal{V}}^X(\mu) \,\mathrm{Det}_\zeta(\Delta_{X,\mathcal{V}} - \mu)$. The determinant comparison formula gives $\widehat{D}_{\mathcal{V}} = C_{\mathcal{V},\mathcal{W}} d_{\mathcal{V},\mathcal{W}}^X \widehat{D}_{\mathcal{W}}$ on $I_{\mathcal{V}} \cap I_{\mathcal{W}}$. Since $d_{\mathcal{V},\mathcal{W}}^X$ and $C_{\mathcal{V},\mathcal{W}}$ are respectively the scalar transition functions of $\mathrm{Det}(\mathcal{E}_X^{\mathbb{R}})$ and $\mathcal{L}_C$, the collection $\{\widehat{D}_{\mathcal{V}}\}$ satisfies the gluing law for local representatives of a smooth section of

$$\mathrm{Det}(\mathcal{E}_X^{\mathbb{R}}) \otimes \mathcal{L}_C.$$

The preceding construction yields the following theorem.

Theorem 6.10. *Let I be a connected component of $\Sigma_X^{\mathbb{R}}$, and let $\mathfrak{A}$ be a collection of Lagrangian pairs such that Assumption 5.1 holds for every pair of realizations indexed by $\mathfrak{A}$ on each connected component of their overlap. For $(\mathcal{V}, \mathcal{V}^{\#}) \in \mathfrak{A}$, set $I_{\mathcal{V}} := I \cap \rho(\Delta_{X,\mathcal{V}})$. Assume that the nonempty sets $I_{\mathcal{V}}$ form an open cover $\mathfrak{U} = \{I_{\mathcal{V}} : (\mathcal{V}, \mathcal{V}^{\#}) \in \mathfrak{A},\ I_{\mathcal{V}} \neq \varnothing\}$ of I, and fix a nonzero frame of $\bigwedge^m \mathcal{V}^{\#}$ for every pair in $\mathfrak{A}$. On every nonempty overlap $I_{\mathcal{V}} \cap I_{\mathcal{W}}$, let $C_{\mathcal{V},\mathcal{W}}$ be the locally constant $\mathbb{C}^\times$-valued function obtained by applying Theorem 5.3 to each connected component of the overlap. Then $\{C_{\mathcal{V},\mathcal{W}}\} \in \check{Z}^1(\mathfrak{U}, \underline{\mathbb{C}^\times})$ and hence determines a rank-one complex local system $\mathcal{L}_C \to I$. Define $\widehat{D}_{\mathcal{V}} : I_{\mathcal{V}} \to \mathbb{C}^\times$ by $\widehat{D}_{\mathcal{V}}(\mu) := E_{\mathcal{V}}^X(\mu) \,\mathrm{Det}_\zeta(\Delta_{X,\mathcal{V}} - \mu)$. Then the collection $\{\widehat{D}_{\mathcal{V}}\}$ defines a smooth section of $\mathrm{Det}(\mathcal{E}_X^{\mathbb{R}}) \otimes \mathcal{L}_C$ over I.*

7. Conclusion

We have constructed the boundary symplectic space of a radial conic Laplacian explicitly from its critical asymptotic coefficients and identified it with the quotient of the maximal domain by the minimal domain. This normalization places the classification of self-adjoint realizations, the Poisson and Weyl operators, Kreĭn's resolvent formula, and the resolvent trace identity in one finite-dimensional boundary framework. Under Assumption 5.1, the same framework gives the positive-spectrum zeta-determinant comparison, including the finite correction from the omitted nonpositive spectrum.

The geometric content is encoded by the holomorphic Weyl curve $\mathcal{K}_X : \Sigma_X \to \mathrm{Gr}_m(V_S)$. The Weyl functions are precisely its local graph coordinates. On the real locus, its tangent form is the $L^2(X')$-pairing of the corresponding formal solutions and is therefore positive. Thus the derivative identity for the Weyl function is the coordinate expression of an intrinsic geometric statement. Likewise, the boundary determinants $d^X_{\mathcal{V},\mathcal{W}}$ are transition functions of the pullback determinant line, while the corrected zeta determinants are local representatives of a section of $\mathrm{Det}(\mathcal{E}^{\mathbb{R}}_X) \otimes \mathcal{L}_C$.

The explicit Fourier analysis used to construct the critical boundary space is specific to the present two-dimensional setting. The standing zeta hypothesis is not asserted here for arbitrary self-adjoint cone realizations. Once an analogous boundary space, Green form, and the required spectral regularity have been established, however, the subsequent Weyl-curve, resolvent, and determinant arguments apply in the same finite-dimensional form.

Acknowledgments

This work was developed over several years and was supported in part by the Ministry of Science and Technology of Taiwan (now the National Science and Technology Council) under Grant Nos. 109-2115-M-006-012, 110-2115-M-006-007, and 111-2115-M-006-018. The author is deeply grateful to his family for their unwavering support during difficult times.

References

[1] D. Burghelea, L. Friedlander and T. Kappeler, Mayer–Vietoris type formula for determinants of elliptic differential operators, J. Funct. Anal. 107 (1992), no. 1, 34–65.

[2] J. Behrndt and M. Langer, Elliptic operators, Dirichlet-to-Neumann maps and quasi boundary triples, in *Operator Methods for Boundary Value Problems*, 121–160, London Math. Soc. Lecture Note Ser., 404, Cambridge University Press, Cambridge, 2012.

[3] J. Behrndt, M. Langer and V. Lotoreichik, Trace formulae and singular values of resolvent power differences of self-adjoint elliptic operators, J. Lond. Math. Soc. (2) 88 (2013), no. 2, 319–337.

[4] J. Cheeger, On the spectral geometry of spaces with cone-like singularities, Proc. Natl. Acad. Sci. USA 76 (1979), no. 5, 2103–2106.
[5] J. Cheeger, Spectral geometry of singular Riemannian spaces, J. Differential Geom. 18 (1983), no. 4, 575–657.
[6] V. A. Derkach and M. M. Malamud, Generalized resolvents and the boundary value problems for Hermitian operators with gaps, J. Funct. Anal. 95 (1991), no. 1, 1–95.
[7] R. Forman, Functional determinants and geometry, Invent. Math. 88 (1987), no. 3, 447–493.
[8] J. B. Gil and G. A. Mendoza, Adjoints of elliptic cone operators, Amer. J. Math. 125 (2003), no. 2, 357–408.
[9] J. B. Gil, T. Krainer and G. A. Mendoza, Geometry and spectra of closed extensions of elliptic cone operators, Canad. J. Math. 59 (2007), no. 4, 742–794.
[10] L. Hillairet and A. Kokotov, Krein formula and S-matrix for Euclidean surfaces with conical singularities, J. Geom. Anal. 23 (2013), no. 3, 1498–1529.
[11] P. Loya, P. McDonald and J. Park, Zeta regularized determinants for conic manifolds, J. Funct. Anal. 242 (2007), no. 1, 195–229.
[12] K. Kirsten, P. Loya and J. Park, Functional determinants for general self-adjoint extensions of Laplace-type operators resulting from the generalized cone, Manuscripta Math. 125 (2008), no. 1, 95–126.
[13] K. Kirsten, P. Loya and J. Park, Exotic expansions and pathological properties of zeta-functions on conic manifolds, with an appendix by B. Vertman, J. Geom. Anal. 18 (2008), no. 3, 835–888.
[14] M. Lesch, *Operators of Fuchs Type, Conical Singularities, and Asymptotic Methods*, Teubner-Texte Math., 136, B. G. Teubner Verlagsgesellschaft mbH, Stuttgart, 1997.
[15] H. Weyl, Über gewöhnliche Differentialgleichungen mit Singularitäten und die zugehörigen Entwicklungen willkürlicher Funktionen, Math. Ann. 68 (1910), no. 2, 220–269.
[16] W. Müller, Relative zeta functions, relative determinants and scattering theory, Comm. Math. Phys. 192 (1998), no. 2, 309–347.
[17] B.-W. Schulze, *Pseudo-Differential Operators on Manifolds with Singularities*, Stud. Math. Appl., 24, North-Holland Publishing Co., Amsterdam, 1991.

[18] S. Scott, Relative zeta determinants and the geometry of the determinant line bundle, Electron. Res. Announc. Amer. Math. Soc. 7 (2001), 8–16.

[19] S. G. Scott and K. P. Wojciechowski, Determinants, Grassmannians and elliptic boundary value problems for the Dirac operator, Lett. Math. Phys. 40 (1997), no. 2, 135–145.

[20] S. G. Scott and K. P. Wojciechowski, The ζ-determinant and Quillen determinant for a Dirac operator on a manifold with boundary, Geom. Funct. Anal. 10 (2000), no. 5, 1202–1236.

[21] J. von Neumann, Allgemeine Eigenwerttheorie Hermitescher Funktionaloperatoren, Math. Ann. 102 (1930), no. 1, 49–131.

[22] Yicao Wang, Complex analysis of symmetric operators, I, arXiv:2408.10968v2 [math.FA], 2024.

Department of Mathematics, National Cheng Kung University, No.1, University Road, Tainan City 701, Taiwan, fjmliou@mail.ncku.edu.tw